Chinook salmon depth distributions on the continental shelf are shaped by interactions between location, season, and individual condition

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Abstract

Background: Ecological and physical conditions vary with depth in aquatic ecosystems, resulting in gradients of habitat suitability. Although variation in vertical distributions among individuals provides evidence of habitat selection, it has been challenging to disentangle how processes at multiple spatio-temporal scales shape behaviour. **Methods**: We collected thousands of observations of depth from > 300 acoustically tagged adult Chinook salmon *Oncorhynchus tshawytscha*, spanning multiple seasons

and years. We used these data to parameterize a machine-learning model to disentangle the influence of spatial, temporal, and dynamic oceanographic variables while accounting for differences in individual condition and maturation stage.

Results: The top performing machine learning model used bathymetric depth ratio (i.e., individual depth relative to seafloor depth) as a response. We found that bathymetry, season, maturation stage, and spatial location most strongly influenced Chinook salmon depth. Chinook salmon bathymetric depth ratios were deepest in shallow water, overwinter, and for immature individuals. We also identified nonlinear interactions among covariates, resulting in spatially-varying effects of zooplankton concentration, lunar cycle, temperature and oxygen concentration.

Conclusions: Our results suggest Chinook salmon vertical habitat use is a function of ecological interactions, not physiological constraints. Temporal and spatial variation in depth distributions could be used to guide management decisions intended to reduce fishery impacts on Chinook salmon. More generally, our findings demonstrate how complex interactions between bathymetry, seasonality, location, and life history stage regulate vertical habitat selection.

Keywords: Habitat use; telemetry; seasonality; ROMS; bathymetry; machine learning models

Introduction

Species exhibit variation in habitat use across spatial and ecological scales. In the case of mobile organisms, habitat use also varies temporally, and individual movement patterns can determine survival and reproductive success [1]. Movement is commonly used to identify processes regulating species distributions, which in turn can improve predictions of how populations will respond to future stressors. For instance, polar bear *Ursus maritimus* habitat use is moderated by sea ice availability [2], while the

migratory patterns of elk *Cervus elaphus* are regulated by interactions between vegetation phenology, quality, and abundance [3].

In the case of aquatic species, however, habitat selection occurs along both horizontal and vertical dimensions. Indeed, environmental gradients often vary more rapidly with depth than with horizontal distance, resulting in discrete species distributions [4]. For example, capelin *Mallotus villosus* are seasonally constrained by subzero temperatures to specific depths in the North Atlantic [5] and species-specific vertical distributions among large pelagic fishes reflect thermal niches [6]. Patterns of vertical habitat use may also highlight difficult-to-observe ecological interactions. Tagging data indicate African penguin *Spheniscus demersus* foraging success varies between benthic and pelagic habitats [7]. The changing climate will likely restructure vertical habitats' ecological characteristics [8, 9]. Thus, understanding the processes shaping depth distributions is necessary to predict how aquatic populations respond to changing environmental conditions.

An improved understanding of how populations use vertical habitats can also directly inform management. Variation in depth can influence a population's vulnerability to fisheries [10] and have been used to tailor fisheries restrictions to minimize impacts on non-target species (e.g., depth-based rockfish conservation areas [11]). Similarly, catchability effects associated with depth can bias indices of abundance and may require modified assessment methods [12]. Fisheries are increasingly regulated by ecosystem-based management, which necessitates an improved understanding of species interactions. Overlapping distributions in both vertical and horizontal space can be used to identify plausible trophic relationships [13, 14].

Unfortunately the ability to predict vertical habitat use is challenged by the diverse processes that influence depth distributions. First, vertical movements consistently differ among habitat types, resulting in patterns across horizontal space [15]. Bottom depth constrains the vertical habitat accessible to an individual, and complex

bottom topographies often serve to aggregate prey thereby impacting predator behaviour [16, 17]. Residual spatial variation in depth distributions may represent physical oceanographic processes, such as offshore currents, which are difficult to observe, but form discrete habitat structures. For example, Porbeagle sharks *Lamna nasus* dive to avoid warm water during migrations [18], whereas billfishes use vertical temperature and oxygen gradients to concentrate prey [19].

Second, vertical movements can vary with time of day and year. Diel vertical migrations (DVM), where individuals follow prey towards surface waters at night, or exhibit the opposite pattern to avoid predators, are particularly widespread [20]. At intermediate temporal scales, changes in the strength of tides, upwelling, or lunar illumination may moderate foraging efficiency or predation risk [16, 6]. Seasonal changes in depth distributions may arise due to physiological constraints, changes in the relative abundance of different prey, or life history events such as reproduction [21, 22, 23].

Third, vertical distribution may consistently differ among individuals. Ontogenetic transitions result in changes in diet, physiological capacity, and foraging behaviour [24, 25]. Depth distributions may change as fish reach reproductive maturity and undergo directed migrations or exhibit spawning behaviours [26, 27]. Individuals often select specific habitats to balance predation risk and foraging requirements [28, 29]. Thus, individuals in poor condition or at greater risk of predation may show distinct vertical distributions.

Importantly, spatial, temporal, and individual processes can interact with one another so that vertical movements are most pronounced under specific conditions. The relative frequency of DVM is moderated by predator communities, prey availability, and individual condition [20]. Blue sharks use ephemeral mesoscale eddies to expand their thermal niche into mesopelagic ecosystems [30]. Maturity stage and habitat influence the vertical distributions of anadromous salmonids—individuals exploit

surface waters for olfactory cues in nearshore habitats, but use deep water thermal refugia during freshwater migrations [31, 32, 33]. Disentangling the relative influence of multiple processes on vertical movements requires observations that span diverse horizontal habitats, multiple seasons, and a large number of individuals. Acoustic tags incorporating pressure sensors provide one opportunity to collect data that meet these criteria. Although the precision of acoustic tags is modest compared to satellite tags, once a network of receivers has been deployed, the cost per tag is lower, which allows for larger sample sizes and a greater capacity to disentangle drivers of individual variation [34].

Chinook salmon *Oncorhynchus tshawytscha* is a piscivorous, anadromous species found in pelagic and neritic habitats in the North Pacific. Chinook salmon support fisheries throughout the region and provide unique ecosystem services, including serving as prey for resident killer whales *Orcinus orca* [35]. A more nuanced understanding of Chinook salmon habitat use can provide novel insights into their ecology and guide management actions. For example, current spatio-temporal closures reflect the two-dimensional habitat use of salmon. Vertical distributions could be used to highlight areas where interactions between Chinook salmon, predators, and prey are most common [13]. Similarly, depth data could be used to minimize interactions between Chinook salmon and midwater trawl fisheries that intercept salmon as bycatch [36, 37]. Typically Chinook salmon occupy depths from the surface to several hundred meters, and variation in vertical habitat use has been attributed to ontogeny, environmental conditions, diel cycles, season, and geographic location [38, 39, 40, 41, 42, 43, 37]. Yet, previous studies have not considered these varied processes simultaneously.

Here we use observations of more than 300 adult Chinook salmon tagged with long-lived depth-sensing acoustic transmitters to identify the processes shaping individual variation in depth distribution. Individual fish were detected using multiple acoustic receivers arrays from the west coast of Vancouver Island (WCVI) to the mouth

of the Columbia River and throughout the Salish Sea (Figure 1). We incorporated an array of explanatory variables that spanned spatial, temporal, and biological processes, including dynamic oceanographic variables from a local Regional Ocean Modeling System (ROMS). We then used a machine learning model to quantify the effect of these covariates on depth and to generate predictions. Ultimately this framework allowed us to estimate how the effect of temporal and biological covariates differed through space, providing novel insights on habitat use in a commercially, culturally, and ecologically important aquatic species.

Methods

Study Species and Area

Chinook salmon are anadromous, semelparous, and have diverse life-history strategies relative to other species of Pacific salmon [44]. After the transition to marine waters, where the majority of growth occurs, Chinook salmon become increasingly piscivorous, feeding on forage fish and squid [45, 44]. We collected observations of Chinook salmon vertical habitat use from northern Vancouver Island to the Columbia River estuary, with the majority of data collected near central Vancouver Island, the coast of southern Washington, Juan de Fuca Strait, and Puget Sound. The study region is bathymetrically diverse, including extensive continental shelf habitats, networks of canyons, narrow straits, and shallow inland seas (Figure 1). Many distinct Chinook salmon populations, originating from central California to northern British Columbia, can be encountered in this region [46, 47]. Some populations use the continental shelf near WCVI and the Salish Sea (defined as Juan de Fuca Strait, the Strait of Georgia, Puget Sound, and their adjoining waterways) as year-round rearing habitats, while others use the region seasonally as a migratory corridor [47].

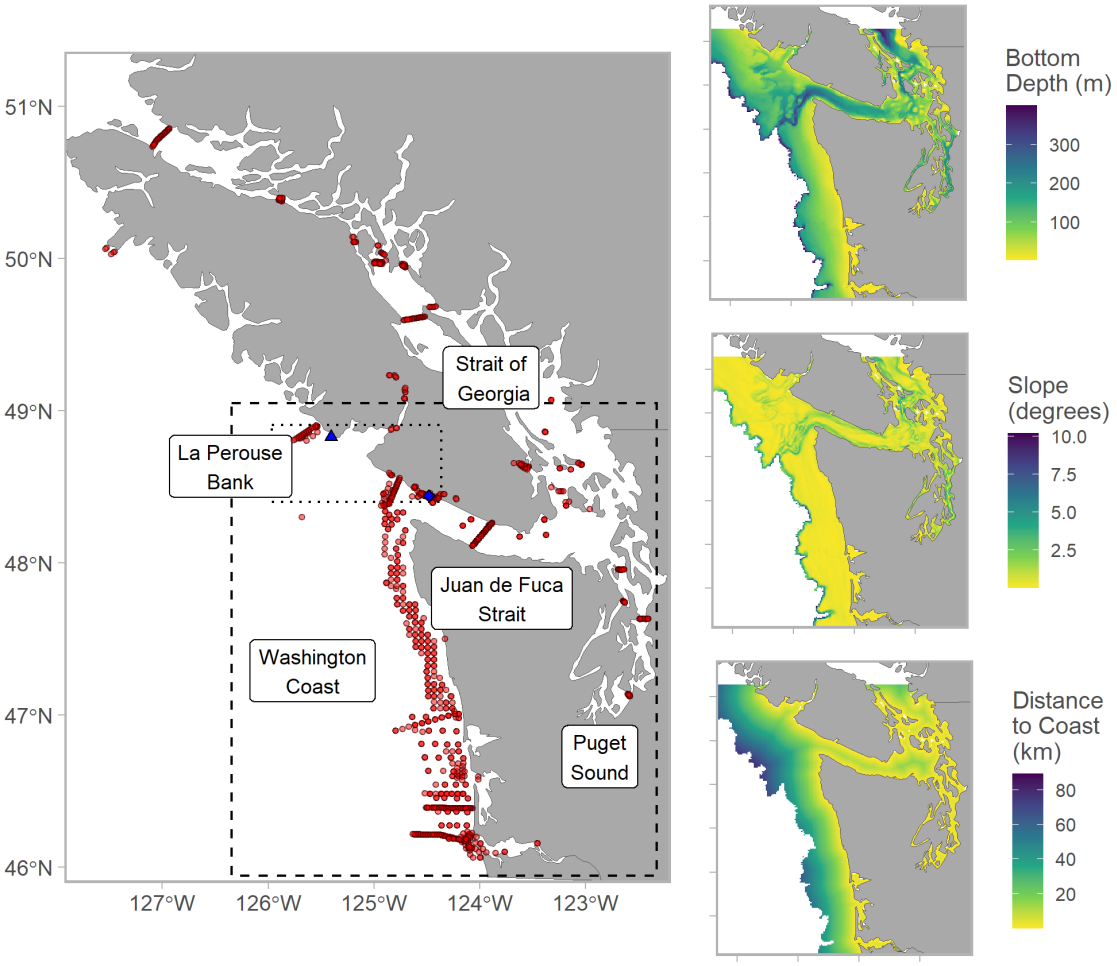


Figure 1: Study area including receiver locations (red circles), as well as tagging locations—Ucluelet (blue triangle) and Port Renfrew (blue diamond). Major geographic features are labelled. Note that the Salish Sea includes Juan de Fuca Strait, the Strait of Georgia, and Puget Sound. The approximate location of tag releases and the focal study area where model predictions were made are shown as rectangles (dotted and dashed respectively). Detailed tag release locations are shown in S1. Column on right shows spatial variables within the study area inset: bottom depth (top), slope (middle) and distance to nearest coastline (right). Projection is UTM Zone 10.

Tag Deployments and Receiver Arrays

Two field programs tagged Chinook salmon. From 2019-2022, fish were sampled by Fisheries and Oceans Canada (DFO) near Ucluelet, British Columbia, using hook-and-line gear fished from a commercial troller. Sampling took place between late April and early September. Fish that were landed without major injuries (e.g., eye damage, heavy bleeding; n = 311) received coded acoustic transmitters (Innovasea Inc.; model V13P; 39 mm length, 5.5 g weight in water, approximately 400 days battery life; mean 120-second delay between transmissions at 69 kHz). Tagging occurred on board the vessel. We transferred fish to a tagging sling with continuous flow-through of ambient seawater. We measured each fish using a tape (fork length and girth), removed a sample from the adipose or caudal fin for genetic stock identification, inserted a passive integrated transponder (PIT) tag, attached the acoustic transmitter, and estimated each individual's energy density using a microwave oscillator, then immediately released the fish. Total handling time never exceeded six minutes. We used Biomark APT12 FDX-B PIT tags and estimated condition using a Distell Model 692 Fish Fatmeter, which provides a non-invasive index of lipid content [48]. We mounted acoustic transmitters externally via a Floy spaghetti tag fixed through the musculature posterior to the dorsal fin. Stock identities were assigned when individual assignment probabilities, estimated using single nucleotide polymorphisms [49], exceeded 80% within a given stock aggregate (Table S1).

A second, University of British Columbia field program deployed acoustic transmitters in 2019 and 2020 (June-August) near Port Renfrew, British Columbia. Fish were sampled using a recreational vessel and sport fishing gear (n = 149), however, the tagging protocol was nearly identical to the DFO protocol, except fish did not receive a PIT tag. Floy spaghetti tags for both DFO and UBC programs included contact information in case a tagged fish was recovered by anglers, hatchery personnel, or biologists conducting spawner surveys. Fish were released between La Perouse Bank

and Juan de Fuca Strait (Figure S1). Both field programs followed protocols approved by institutional animal care committees (DFO AUP 19-011, UBC AUP A19-0193).

We detected tagged fish on Innovasea acoustic receivers (VR2, VR3, and VR4 models) that were deployed throughout the study area. These arrays were managed by a diverse network of scientists and the configuration of the arrays differed among years (Figure S7). Details in Online Supplement.

Explanatory Variables

We quantified relationships between depth and explanatory variables associated with detections (spatial location, time, oceanography) or the characteristics of individual fish at tagging (Table 1). We also considered a model that included stock aggregate identity as an additional covariate (Table S1).

Spatial Variables

We included the location of each detection (UTM coordinates in zone 10) and the distance to the nearest shoreline. We also included two bathymetric variables—bottom depth and bottom slope. Since the precise location of a detection within a receiver's detection radius is unknown, we calculated each receiver's mean bottom depth and mean slope within an 800 m radius. Bathymetric data were downloaded from NOAA's 3-arc second (British Columbia; [50]) or 1/3-arc second (coastal Washington; [51]) digital elevation models. By including spatial coordinates as covariates, we developed a model that accounted for residual spatial variation that was present after spatially correlated variables, were incorporated.

Temporal Variables

We included calendar day of detection, which was transformed to represent seasonal effects. Details in Online Supplement. To account for lunar effects, we calculated the

Table 1: Explanatory variables included in statistical analysis. Names in the second column match Figure 3.

Group	Covariate	Definition
Spatial	Location (UTM X and Y)	Easting and northing of receiver in UTM zone 10
	Shoreline Distance	Distance (km) to coastline
	Bottom Depth	Mean depth (m) of seafloor
	Bottom Slope	Mean slope (°) of seafloor
Temporal	Year Day (1 and 2)	Calendar day converted to two variables representing cyclical process
	Lunar Cycle	Proportion of moon illuminated (0-1)
	Day-Night	Before/after sunset
Oceanographic	Temperature	Sea surface temperature (C°)
	Hor. Current 1	Horizontal momentum (m/s) in N-S direction
	Hor. Current 2	Horizontal momentum (m/s) in E-W direction
	Vert. Current	Vertical momentum (m/s)
	Zooplankton	Zooplankton concentration ($\frac{\text{mmol N}}{\text{m}^3}$)
	Oxygen	Dissolved oxygen concentration ($\frac{\text{mmol O}}{\text{m}^3}$)
	Thermocline Depth	Mixed layer depth (m) based on vertical temperature profiles
Individual	Size	Fork length when tagged (cm)
	Lipid Content	Converted whole body lipid content (% wet weight)
	Maturity	Whether individual will return to spawn in the same calendar year as the detection event

proportion of the moon's face that was illuminated at a given date and geographic location using the oce R package [52]. To account for diurnal effects, we created a categorical variable representing whether a detection event occurred during or after daylight hours (defined as sunrise and sunset), based on local time and the spatial location of the detection, using the suncalc R package [53].

Oceanographic Variables

We obtained covariates representing dynamic oceanographic conditions from the LiveOcean configuration [54] of the Regional Ocean Modeling System (ROMS) [55, 56]. ROMS is a free-surface, primitive equations ocean circulation model that simulates ocean responses to physical forcing by wind, heat, tides, and other drivers. All LiveOcean variables were extracted at the surface (preliminary results were qualitatively similar with 25 m depth). We calculated thermocline depth using the rLakeAnalyzer R package [57], based on vertical profiles of temperature extracted at each detection location. We used temperature and oxygen concentration estimates at depth for each detection to visualize habitat use. Details in Online Supplement.

Individual Variables

We included individual measurements of fork length and an index of lipid content to account for differences in individual condition at the time of tagging. We used a Distell Model 692 Fish Fatmeter (Distell Inc., West Lothian, Scotland), to generate indices of tissue lipid concentration. The Fatmeter uses a microwave oscillator to emit a low-powered wave (frequency, 2 GHz 6 2,000 MHz; power, 2 mW) that interacts with water in the somatic tissues at a given body location. We took Fatmeter readings at two positions anterior of the dorsal fine and above the lateral line [48]. We averaged these data to generate a single index, which was converted to an estimate of whole body lipid content following methods outlined in [58].

We could not infer maturity stage at time of capture because Chinook salmon differ in maturation age and did not show external signs of sexual maturity. Therefore we assigned maturity stage post-hoc in a two-step process. In the first step, all fish detected in-river (via acoustic telemetry, PIT arrays, harvest, escapement walks, or hatchery broodstock removals) in the year of tagging were classified as mature, and all fish that were detected on marine arrays after November 15th in the year of tagging were classified as immature. In the second step, the remaining individuals (approximately one-third of the total) were assigned to a maturity stage based on size at the time of capture and capture date. Briefly, we fit a logistic regression model to fish of known maturity stage, including fork length and capture date as fixed effects and a random intercept for stock identity. We used the estimated model parameters to identify stock-specific sizes and capture dates at which the probability of being mature exceeded 50%. Fish with an uncertain maturity stage based on detections were assigned to a maturity stage based on this population-specific threshold size. Immature fish were considered mature on May 1 of the year following tagging. Details in Online Supplement.

Statistical Modeling

Depth data collected from acoustic transmitters have several characteristics that complicate parametric analyses. First, detection data are serially autocorrelated with substantial temporal and spatial structure. Second, depth is not normally distributed but bounded by zero and, in this case, right skewed towards surface depths. Third, depth distributions are constrained by bottom depth, resulting in a truncated distribution that varies spatially. As a result, parametric models predicting depth often exhibit heteroskedastic residuals and unrealistic predictions.

As an alternative, we used machine learning algorithms to estimate the relative importance of covariates on detection depth, find non-linear and interactive effects on

detection depth, and generate spatially explicit predictions. For brevity, we describe one algorithm, random forest regressions [59], in detail here, but we also evaluated gradient boosting machines ([60]; see Online Supplement for details). Random forest regression algorithms are built on regression trees. Each regression tree splits observations along a single covariate at the point minimizing the sum of squares error, where the prediction at each node is the mean of the response variable. This process is repeated, sequentially selecting covariates and split points, until each terminal node contains less than a specified number of observations. Since individual trees are sensitive to overfitting and generally poor at prediction, random forest algorithms average predictions across regression trees. Individual regression trees within random forest models are fit to a bootstrap sample of data and the models can be tuned with out-of-sample cross validation, making these models robust to temporal and spatial autocorrelation, and allowing them to develop an appropriate level of parsimony. Thus we were able to include all valid detections without binning data and we could explore interactions among many covariates. Machine learning models also do not assume the response variable follows an underlying statistical distribution, increasing flexibility when modeling data with a non-standard truncated distribution.

We tested both machine learning algorithms, as well as alternative suites of hyperparameters that control each algorithm's learning process. We compared models and alternative hyperparameters using 8-fold cross-validation. We blocked training and testing data by individual fish to account for differences in migratory behaviour, as well as interannual differences in the location of receivers. Model selection and hyperparameter tuning excluded tags deployed in 2022 and randomly assigned data to training or testing groups based on block. For each model, we evaluated predicted performance using three alternative transformations for observed depth data: untransformed (bounded by zero and bottom depth within a detection radius), a bathymetric depth ratio (observed depth divided by the maximum depth within a receiver's de-

267 tection radius; bounded by zero and one), and a logit transform of the bathymetric
268 depth ratio (approximately normally distributed). We compared alternative model
269 structures using root mean square error (predictions based on transformed response
270 variables were back-transformed). We also tested predictions from our 2019-2021 data
271 model 2022 deployment data as an independent test of future predictive performance.
272 Additional details in Online Supplement.

273 We fit machine learning models using the caret [61] and ranger [62] R packages.
274 We interpolated a small number of missing ROMS and lipid content estimates (details
275 in Online Supplement) and transformed categorical variables (day-night and maturity
276 stage) to dummy variables prior to model fitting. We quantified the relative explana-
277 tory power of different variables by calculating the decline in model performance (per-
278 cent mean squared error) when a given covariate was excluded from the model (i.e.,
279 the performance of trees that included a given variable relative to those that did not).
280 We generated a suite of conditional predictions to evaluate the effect of individual
281 covariates and better understand interactions among covariates. Predictions included
282 a) non-spatial conditional predictions that focused on the effect of single covariates
283 with all other covariates (including spatial location) held at reference values (median
284 for continuous and 0.5 for dummy variables) and b) conditional spatial predictions
285 that used a grid of cells (1×1 km resolution) with relevant spatial attributes (bottom
286 depth, slope, distance to shore) and all other non-focal variables fixed at reference
287 values (median values except for year day fixed to July 30, dynamic ocean variables
288 fixed to estimates for July 30 2020, and thermocline depth fixed at its mean value for
289 July). Since machine learning models capture non-linear relationships among large
290 numbers of covariates, we also generated conditional spatial contrasts by calculating
291 the difference in spatially explicit predictions, which allowed us to visualize spatially
292 varying effects.

To quantify uncertainty, we generated infinitesimal jackknife confidence intervals [63] and among-tree quantile prediction intervals using the ranger package [62]. All analyses were completed in R 4.2.1 [64].

Results

Immature fish were smaller and had a lower lipid content (67.0 ± 5.51 cm and 7.4 ± 1.23 % wet weight; n = 48) than mature fish (78.7 ± 8.78 cm and 8.1 ± 1.67 % wet weight; n = 412). Fork length, lipid content, and the relative proportion of immature fish varied among stocks (Figure S2).

We used 43,857 detections with valid depth data from 339 Chinook salmon tagged between 2019 and 2021 to train and test the depth model using cross validation. An additional 5,498 detections from 17 individuals tagged in 2022 were used to qualitatively evaluate model performance. The number of detections per tag ranged from 1 to 5,852 and tags provided detection data for up to 437 days (Figure S3). Detection depth of Chinook salmon (hereafter depth) ranged from the surface to 352 m, with a median of 28 m and deeper detections among immature fish (Figure S4). Chinook salmon were detected in both surface waters and along the sea floor (Figure S4). The number of detections and mean depth of detections varied among receivers (Figure S7, S8). Chinook salmon moved through a wide range of temperatures (6-18 C°; Figure S5) and dissolved oxygen concentrations (<1-12 mg/l, Figure S6).

The best-supported machine learning model was a random forest regression model fit to bathymetric depth ratio (i.e., observed depth scaled by the maximum bottom depth within the receiver's detection radius; Online Supplement). The model included 1500 trees, evaluated 13 variables per tree, and identified split points randomly following a uniform distribution (i.e., extremely randomised trees [65]). The model had unbiased predictive performance with in- and out-of-sample data and high predictive

accuracy with in-sample data (r^2 = 0.83); however, root mean square error increased with out-of-sample data, particularly with data associated with detections from 2022 tag deployments (Figure 2).

The strongest predictors of Chinook salmon depth distribution were bottom depth, one of two composite calendar day variables, maturity stage, and location. The remaining temporal, spatial, and individual variables, as well as a subset of oceanographic variables, had more modest contributions. Current strength and thermocline depth had the poorest predictive performance (Figure 3). Stock identity covariates also had weak predictive power (Figure S9), and the remaining results focus on the model that excluded stock identity effects to improve interpretability.

Generally Chinook salmon were predicted to occupy the top 20-40% of the water column, relative to bottom bathymetry (Figures 4). The cumulative influence of bottom depth, spatial location, bottom slope, and distance to shore resulted in strong spatial patterns in vertical distribution. Bathymetric depth ratios were shallowest in western Juan de Fuca Strait, the southern Strait of Georgia, and Puget Sound (Figure 4). Predicted depth was markedly deeper—50-60% of water column depth—on shelf habitats off the Washington coast and portions of coastal WCVI (Figure 4). Back-transformed predictions (i.e., depth in meters) were deepest in the southern Strait of Georgia and in the canyons near the western portion of Juan de Fuca Strait (Figure 4).

Many of the covariates we considered had nonlinear effects on depth. Chinook salmon were more likely to occupy depths closer to the bottom in shallower waters and when bottom topography was relatively flat (Figures 5, S4). We found evidence of cyclical, seasonal changes in Chinook salmon depth, where fish were deeper during winter months (Figure 5). Increases in mean depth were rapid, occurring over one to two weeks during April (when fish moved to shallower waters) and September (when they returned to deeper waters). Immature fish tended to have shallower distributions than mature fish, particularly after accounting for correlations between

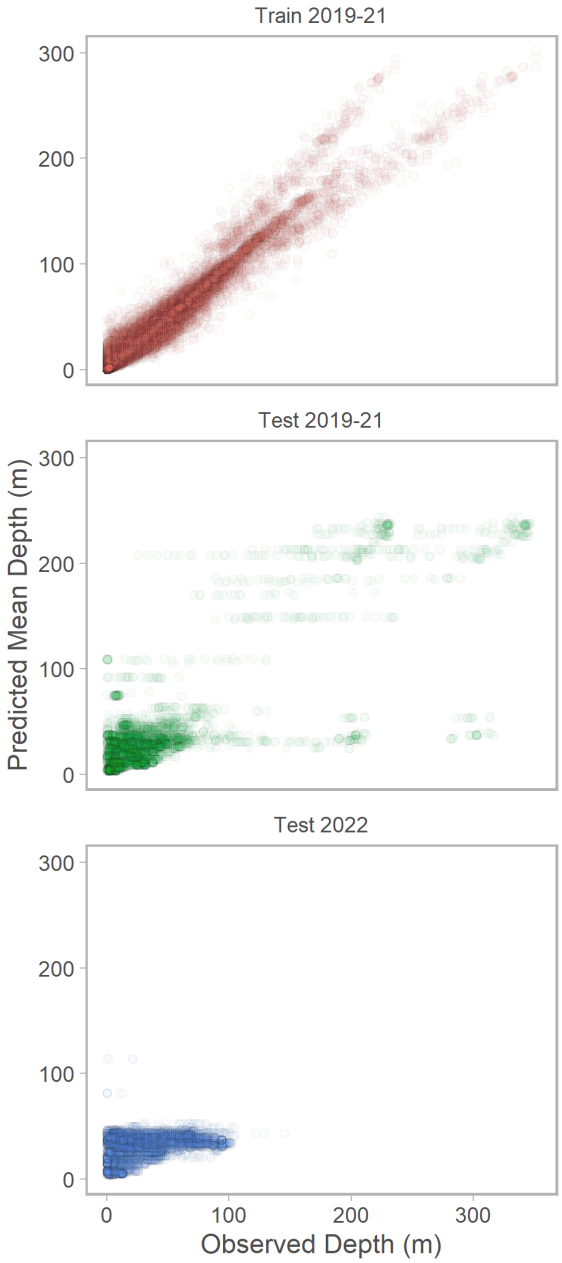


Figure 2: Observed and predicted depth showing 2019-21 training data from 8-fold cross validation used to fit the initial model (red; 80.1% of detections), 2019-21 out-of-sample testing data from 8-fold cross validation (green; 8.8% of detections), and out-of-sample testing data from 2022 tag deployments (blue; 11.1% of detections).

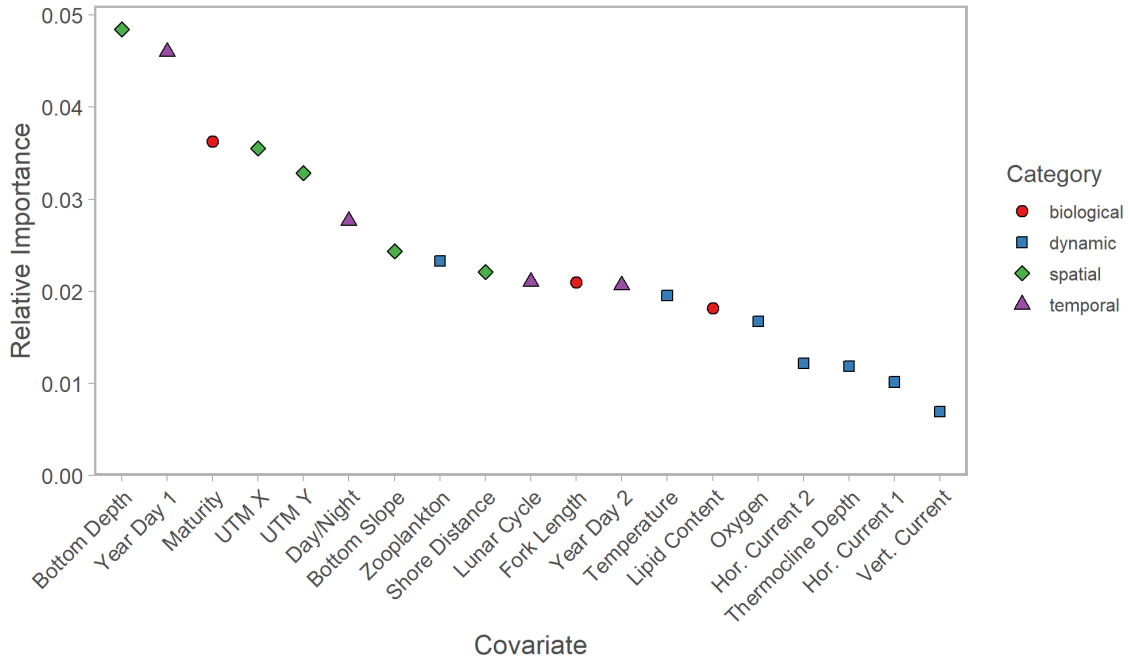


Figure 3: Relative importance of explanatory variables in random forest model. Importance is quantified as the difference (mean among trees) in root mean square error when a given variable is excluded during out-of-bag prediction. An uncorrelated random variable will have a difference of zero. Colors represent categories of explanatory variables.

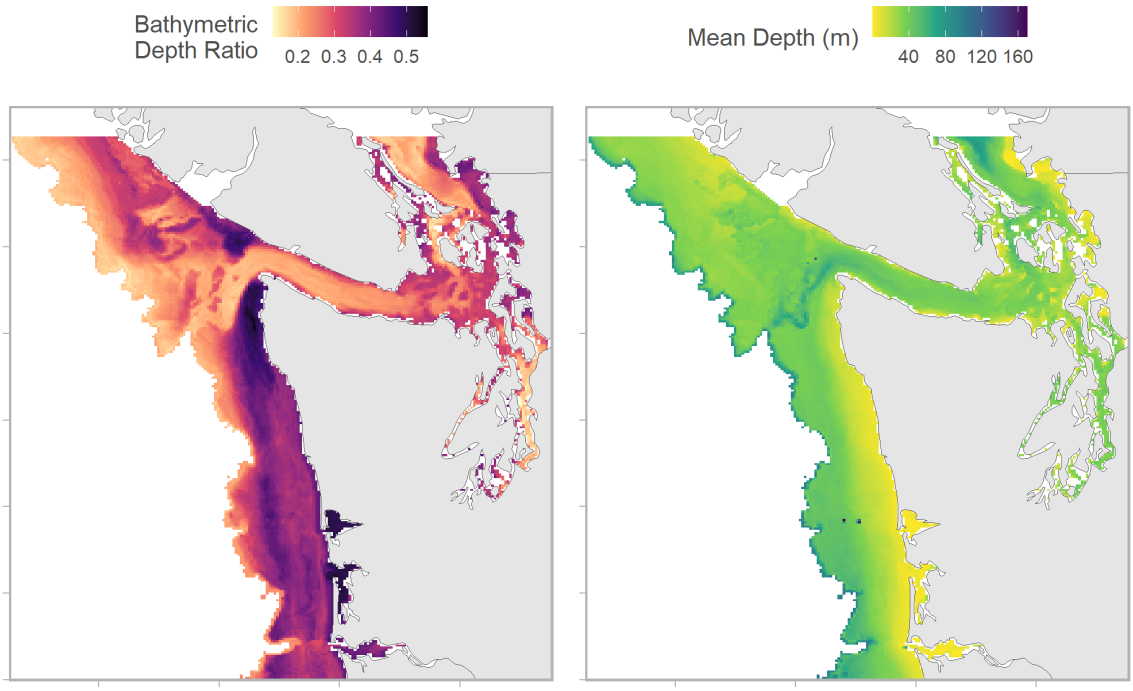


Figure 4: Model predicted mean depth (left) and bathymetric depth ratio (right) of Chinook salmon. For bathymetric depth ratio, zero represents the surface and one the seafloor. Predictions are conditional effects of spatial processes, with year day fixed to 211 (i.e. July 30) and dynamic oceanographic variables fixed to estimated values for July 30, 2020. All other temporal and biological variables are fixed at median values or 0.5 for dummy variables.

maturation stage, size, and lipid content, and nocturnal predictions were shallower than diurnal (Figure 5). Finally, after accounting for the effect of other spatial covariates (e.g., bottom depth, slope, distance to shore, dynamic oceanographic variables), bathymetric depth ratio deepened in southern and, to a lesser extent, eastern locations relative to southwest Vancouver Island (Figure 5).

The effect of the remaining covariates was generally modest—less than 5% over the range of the observed covariate. However, considering covariates in isolation can obscure interactions between spatial processes and other variables that result in stronger localized effects. For example, the distributions of Chinook salmon deepened when zooplankton concentration, lunar illumination, sea surface temperature, and oxygen concentration increased (Figure 6). With the exception of lunar illumination, these effects were constrained to portions of the continental shelf.

Discussion

Variation in individual depths provides information on the interaction between multiple ecological processes that regulate habitat use in marine species. We used observations from more than 300 adult Chinook salmon tagged during a four-year period to disentangle the effect of spatial, temporal, physical, and biological processes on vertical distributions. Tagged individuals ranged from the surface to nearly 400 m in depth, but were typically distributed near the middle of the water column approximately 25 m below the surface. Chinook salmon encountered a wide range of thermal habitats, from six to 18 C°, and intermittently hypoxic conditions.

Chinook salmon depth distributions were most strongly correlated with bottom depth and season. As bottom depth increased, fish shifted from being located in the center of the water column to the top quarter, with the steepest change in vertical distribution occurring at bottom depths up to 100 m. After accounting for the effect

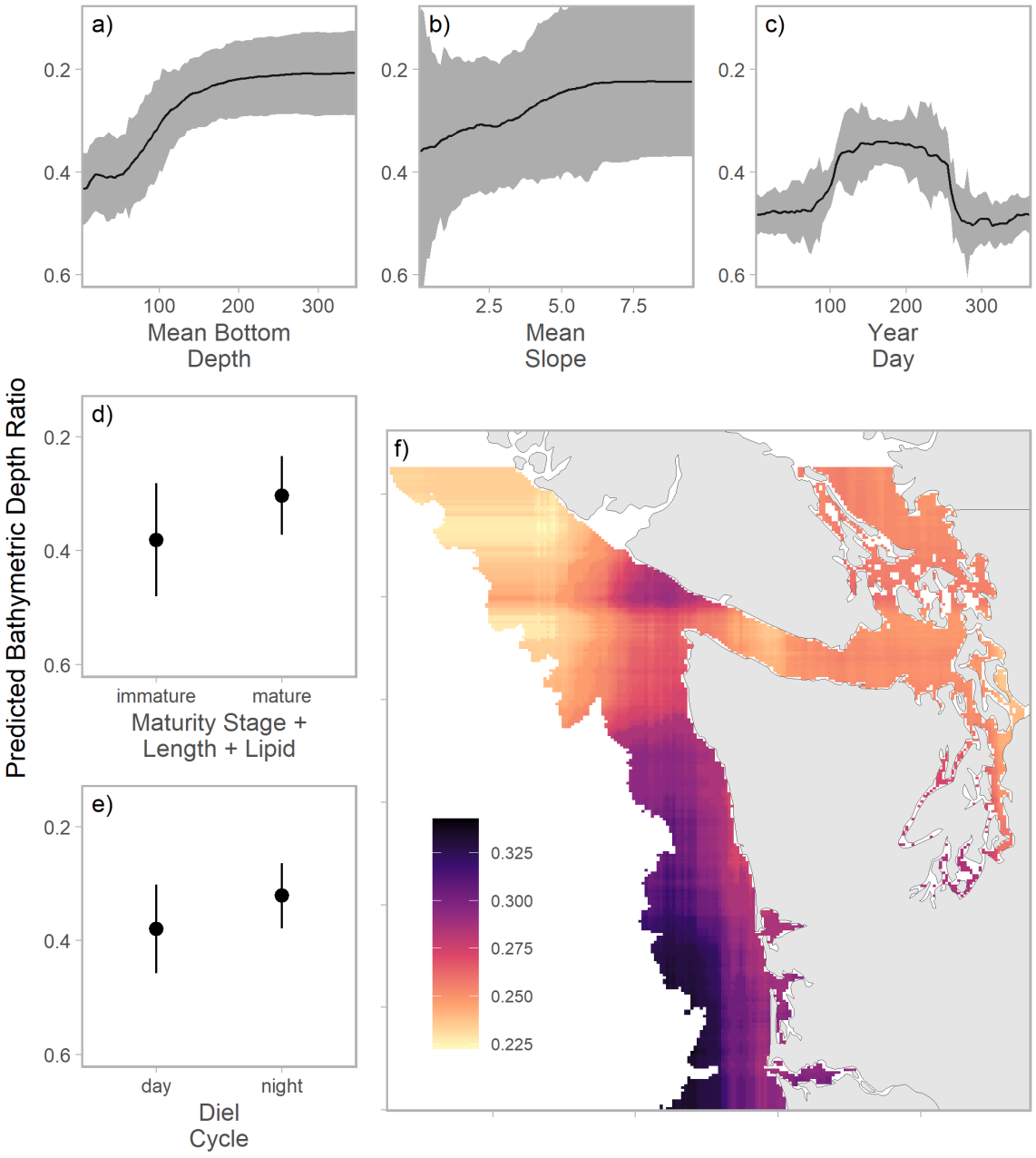


Figure 5: Conditional effects on Chinook salmon bathymetric depth ratio (zero at surface and one at sea floor) of: bottom depth (a), bottom slope (b), day of detection (c), maturity stage/fork length/lipid content (d), day/night (e), and easting-northing (f). Conditional effects represent the mean predictions (lines or point) assuming all other variables are fixed to reference values (i.e., medians for continuous covariates and 0.5 for dummy variables). Predictions based on maturity stage assume stage-specific mean values for fork length and lipid content. Predictions based on location exclude the effect of spatially correlated variables (i.e., mean bottom depth, mean bottom slope, distance to shore, dynamic oceanographic features). Ribbons and whiskers represent 90% jackknife confidence intervals.

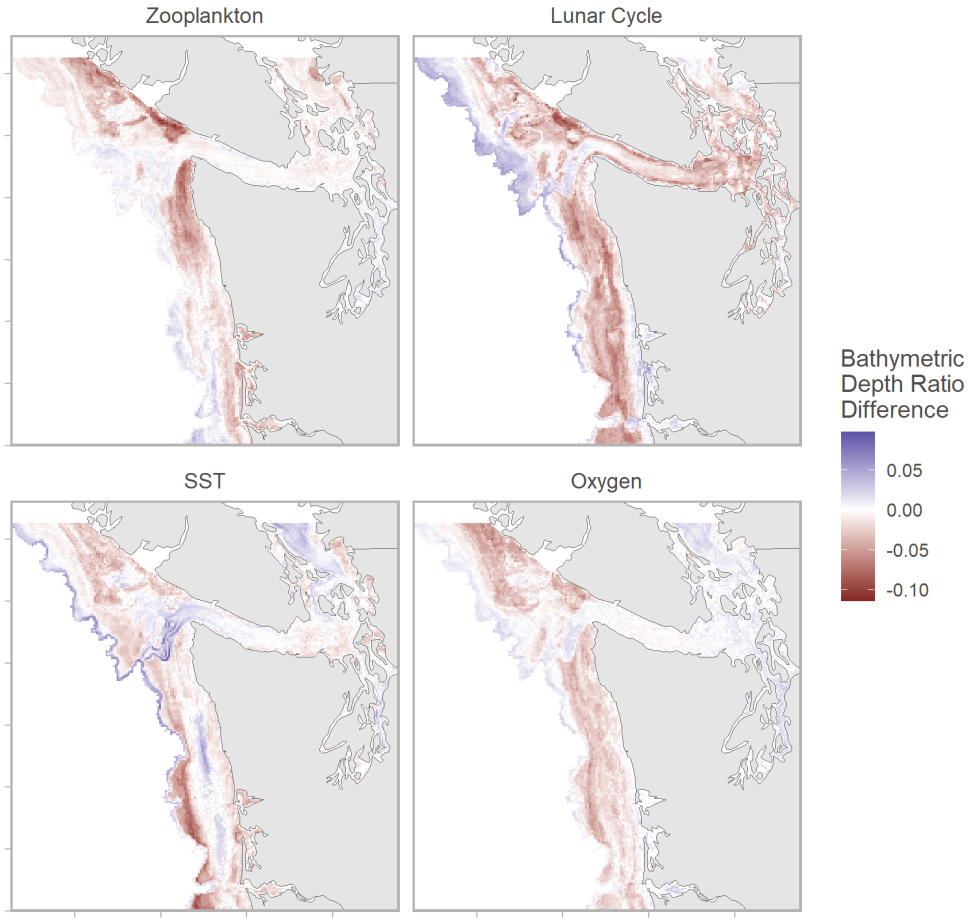


Figure 6: Model predicted conditional differences in bathymetric depth ratio across the study area. Maps show predicted differences when: sea surface zooplankton concentrations are 1 SD below or above their summer average (top left); the moon is 0% and 100% illuminated (top right); sea surface temperatures are 1 SD below or above their summer average (bottom left); and sea surface oxygen concentrations are 1 SD below or above their summer average (bottom right). Relative differences were calculated as the difference between groups (e.g., 0% and 100% illumination) in a given spatial cell. Negative (red) values represent greater predicted depth with greater lunar illumination, temperature, zooplankton concentration, or oxygen concentration. Non-focal spatial covariates were fixed at July 30, 2020 values and the remaining covariates were fixed at median values.

of maturation stage, Chinook salmon depths increased over winter. Seasonal transitions were relatively rapid and occurred over one to two-week periods in April and September. We also identified significant residual spatial structure in Chinook salmon vertical distributions. In other words, there were distinct locations where detection depth varied independently of bottom depth, slope, distance to shore, or dynamic oceanographic features. This spatial effect resulted in relatively shallow distributions in the northwest portion of the domain, deep distributions along the southern continental shelf, and intermediate distributions in Juan de Fuca Strait and the southern Strait of Georgia.

After accounting for seasonal and spatial effects, we found relatively strong effects of maturation stage and diel cycle. Although fork length and lipid content at the time of tagging had weaker predictive power, they were correlated with maturation stage and we estimated maturation stage effects fixing these covariates at their stage-specific mean values. Depth distributions were typically shallower for mature individuals and overnight.

Dynamic oceanographic conditions and lunar cycle had weaker effects on vertical distribution. Yet, each variable still provided information that improved model performance. Furthermore, the flexibility of machine learning models allowed us to consider interactions among many variables, such as spatially varying effects. These analyses indicated that several variables with relatively weak overall effects influenced vertical distributions in specific locations. For example, Chinook salmon in eastern Juan de Fuca Strait and along the continental shelf showed deeper distributions when the moon was fully illuminated. Similarly, the effect of zooplankton and oxygen concentrations on Chinook salmon depth was negligible in the Salish Sea but relatively strong along the continental shelf. Sea surface temperature effects were particularly heterogeneous with Chinook salmon showing deeper distributions when temperatures were

warm in regions with shallow bathymetries, but the opposite relationship in regions with deeper bottom depths.

Ecological Drivers of Depth Distribution

We predict that the primary drivers of variability in Chinook salmon depth distributions are prey availability and associated foraging behaviours. A position within the middle of the water column is consistent with a vertical ambush strategy whereby larger-bodied piscivores attack pelagic prey from below [66]. Moreover, several of the top-ranked covariates have clear linkages to prey availability. Complex bottom bathymetries increase bio-physical coupling as nutrients are deflected from deeper waters, resulting in greater primary productivity and concentrating zooplankton and forage fish [67, 16]. In this study, bottom depth and slope were correlated with individual depth, and also with one another, suggesting these variables are proxies for bottom complexity. Furthermore, steep dropoffs, reefs, or pinnacles on the west coast of Vancouver Island often have greater Chinook salmon abundance and are commonly targeted by fishers (C. Freshwater and B. Hendriks, unpublished data). In this paradigm, residual spatial variation likely represents foraging hotspots associated with variables excluded from the model (e.g., bottom substrate, locations of tidal outflow).

Seasonal changes in Chinook salmon depths mirror changes in prey species. Forage fish in the northeast Pacific, particularly Pacific herring *Clupea pallasii* and northern anchovy *Engraulis mordax*, become increasingly bottom-oriented during the winter [68, 69, 70] and are key prey of Chinook salmon [44, 45]. The correlation between zooplankton concentration and depth, though localized, suggests Chinook salmon behaviour covaries with productivity in rearing habitats.

Although we believe our results are most consistent with foraging behaviour, Chinook salmon may also moderate their depth in response to predation risk. Resident killer whales prey upon Chinook salmon throughout the year [71, 72] and have similar

depth distributions [13]. Tagged Chinook salmon exhibited deeper distributions relative to bottom bathymetry near western Juan de Fuca Strait and the southern Strait of Georgia, which have been identified as resident killer whale foraging hotspots [73]. Both cetacean and pinniped habitat use is also correlated with bottom depth and topography [74, 75], presumably because these features concentrate prey. Moreover, the depth of salmon sharks *Lamna ditropis*, another salmon predator, changes seasonally [76], suggesting overwinter changes in depth may have knock-on effects across trophic levels.

The distributions of mature Pacific salmon shallow as they use olfactory cues to home to natal streams [31, 32]. As a result, we anticipated Chinook salmon distributions would vary with ontogeny and along their migration corridor, resulting in shallower distributions for mature individuals and as fish approached terminal areas. Immature Chinook salmon, which were also smaller and had a lower lipid content than mature individuals, did have deeper distributions, even after accounting for seasonal effects. The spatially varying effects of zooplankton concentration we observed were also consistent with fish responding more strongly to productivity in rearing areas, such as La Perouse Bank and coastal Washington, than in migratory locations such as the southern Strait of Georgia. Notably, stock identity was a poor predictor of variation in depth distribution, presumably because other model covariates accounted for stock-specific traits such as body size and migration route.

We did not find strong evidence that oceanographic conditions regulated Chinook salmon distributions. Zooplankton concentration, sea surface temperature, and oxygen concentration had modest performance, but their predictive power was substantially weaker than other covariates, and the effects of thermocline depth and current velocity were weaker still. Consistent with previous Pacific salmon tagging studies, we found tagged individuals encountered a wide range of environmental conditions

[33, 43]. Chinook salmon may adopt strategies similar to other pelagic piscivores, whereby physiologically taxing habitats are used to concentrate prey [66, 19].

Divergent Patterns Across Aquatic Habitats

While we found spatial and seasonal variables were the strongest drivers of variation in depth distribution, previous work on Pacific salmon and other large pelagic fishes has emphasized the importance of processes occurring at shorter temporal scales. We believe that our divergent results are due to the ontogeny of tagged fish and their habitats. Previous work on Pacific salmon has demonstrated that vertical and horizontal habitat use is moderated by temperature and oxygen. Many of these studies have focused on individuals that have ceased feeding to undergo terminal migrations. Individuals in terminal areas use surface waters for orientation [31, 32] and deeper habitats as thermal refugia [77, 33, 78]. Conversely, the Chinook salmon we tagged were continuing to forage and were months away from entering freshwater and may have been less likely to exhibit strong responses to physical gradients [41, 43].

While there is evidence of temperature or oxygen constraining Chinook salmon habitat use in California Current habitats [79, 37], temperatures in this region are considerably warmer than those off southern Vancouver Island and northern Washington and oxygen may be more limiting due to stronger seasonal upwelling [4, 80]. Indeed, Sabal et al. 2023 reported that a) the warmest temperatures they reported were not observed in the northern portion of their domain (which overlapped with our study area) and b) the Chinook salmon populations observed in northern regions did not show strong responses to temperature [37]. We found that Chinook salmon showed moderate responses to sea surface temperature and oxygen concentration in shelf habitats and were consistently deeper in the water column on the Washington continental shelf than elsewhere. Both patterns suggest that the effects of dynamic oceanographic conditions vary spatially.

More generally, our tagging data can be used to compare the vertical movements of Chinook salmon to well-studied species such as billfishes, tunas, and sharks (hereafter large pelagics). Chinook salmon and large pelagics are in many ways ecologically similar. They are predominantly piscivorous, undergo extensive migrations, and often exhibit diving behaviour. Large pelagics such as dolphinfish *Coryphaena hippurus* and silky sharks *Carcharhinus falciformis* are largely constrained to the epipelagic. Blue sharks *Prionace glauca* and bigeye tuna *Thunnus obesus*, however, dive into the mesopelagic where they encounter cooler temperatures and hypoxic conditions [6]. Similar to the latter species, we found Chinook salmon moved through a wide range of temperatures, intermittently co-occurred with very low oxygen concentrations, and were not constrained by the thermocline. Like some large pelagics [6], Chinook salmon were shallower at night and distributed deeper in the water column when light levels increased, presumably due to improved foraging efficiency at depth. Although the effect of diel period was weaker than many other covariates, the forage fish that Chinook salmon prey upon do migrate to shallower depths at night [81], which may elicit a behavioural response particularly prior to directed migrations.

Unlike large pelagics, however, Chinook salmon were not strongly associated with thermocline depth or horizontal and vertical currents, a proxy for mesoscale features [30]. Again we suggest that habitat characteristics drive these differences. Tagging studies focused on large pelagics typically occur in offshore regions with substantial stratification [10, 6], a more extreme version of the California Current Chinook salmon studies described above. Our study area includes regions that are relatively well-mixed due to estuarine circulation (Juan de Fuca Strait) or seasonal upwelling (coastal Washington and WCVI) and generally shows less vertical structure in temperature than tropical regions [4].

Limitations and Conclusions

Our analysis of Chinook salmon depth distributions using acoustic telemetry data has several limitations. First, we did not estimate horizontal and vertical habitat use simultaneously. A shift in Chinook salmon distributions from migratory corridors to productive habitats, coincident with increases in depth, would be strong evidence that foraging decisions moderate seasonal variation; however, developing fully three-dimensional models was outside the scope of this work. Second, our estimates of vertical habitat use include several sources of uncertainty. Since fish locations could not be identified within the detection radius of a receiver, our model integrated variation within that radius and could obscure fine-scale differences in habitat use. While ROMS model outputs accurately capture relatively large-scale variability in oceanographic conditions, they likely do not fully reflect granular environmental processes that serve as cues for individual fish. Size and lipid content at the time of tagging are reasonable short-term proxies for body condition, but our model did not account for individual growth. Size, lipid content, and other traits also differ among Chinook salmon stocks. While including stock identity did not improve model performance, it remains unclear to what extent differences among Chinook salmon are due to individual variation as opposed to stock-specific behaviours that covary with traits such as size or migration route. Third, the populations we tagged, and their associated habitats, represent only a fraction of the ecological diversity within Chinook salmon. Chinook salmon behaviour may vary regionally or among life-history types that were not well sampled here (e.g., populations with yearling life histories that mature offshore). More generally, we could only sample depth distributions in locations and times where receivers were deployed. The accuracy of our model's predictions will be reduced if habitat use differs in unmonitored locations or times. Indeed, model performance worsened when data from 2022 tag deployments were compared to predictions from the model trained on 2019-21 tag deployment data.

Unlike open ocean pelagic environments, we found that vertical habitat use on the continental shelf was moderated at relatively coarse scales by interactions between static spatial variables and seasonality. Processes occurring over shorter temporal periods had weaker or more localized effects. Variation in Chinook salmon depth distributions provides both challenges and opportunities for management. On the one hand, strong seasonal effects coupled with a broad vertical distribution suggest catchability varies, complicating the interpretation of fisheries-dependent data [12]. On the other, evidence that depth varies spatially, seasonally, and ontogenetically could guide interventions intended to constrain directed and incidental harvest. Previous work has highlighted how trade-offs between growth, predation risk, and reproductive success can influence habitat selection in aquatic species [28, 82]. Our analysis provides a framework for extending this work into difficult-to-observe pelagic ecosystems by linking tagging data, dynamic oceanographic models, and spatially explicit models.

Declarations

Availability of Data and Materials

The data and code underpinning these analyses are available from https://github.com/pacific-salmon-assess/chin_depth.

Competing Interests

The authors declare that they have no competing interests.

Funding

This work was supported by Ecosystem Science Division, Fisheries and Oceans Canada, which funded DFO field work; the British Columbia Salmon Restoration and Innovation Fund, which funded UBC field work; and the U.S. Navy, U.S. Pacific Fleet, Environmental Readiness Division, which provided funding for the receiver array along the US Washington Coast.

Author Contributions

CF designed the study with input from SH, AT, and JK. CF, SA, and JK developed the methodology. CF, DH,JS, DJ, BH, and SJ led data collection. CF and SA led the data analysis. CF drafted the original version of the manuscript. All authors contributed to revising the manuscript and approved the final draft.

Acknowledgements

We thank Erika Nielsen, Katie Innes, Kelsey Flynn, Willem Offerein, and Simon Offerein for assistance with tag deployments. We also thank Peter van Buren, Caitlin O'Neal, David Welch, Aswea Porter, Megan Moore, Barry Berejikian, Ocean Networks Canada, and Ocean Tracking Network, who were integral in deploying receiver arrays and in sharing data. We are grateful for constructive feedback on a previous version of this manuscript from Megan Moore, Aswea Porter, and Michelle Charbonneau.

References

[1] Nathan R, Getz WM, Revilla E, Holyoak M, Kadmon R, Saltz D, et al. A movement ecology paradigm for unifying organismal movement research. 2008. Proc Natl Acad Sci USA;105(49):19052-9.

[2] Ferguson SH, Taylor MK, Messier F. Influence of sea ice dynamics on habitat selection by polar bears. 2000. Ecology;81(3):761-72.

[3] Hebblewhite M, Merrill E, McDermid G. A multi-scale test of the forage maturation hypothesis in a partially migratory ungulate population. 2008. Ecol Mono;78(2):141-66.

[4] Thomson RE. Oceanography of the British Columbia Coast. 1981. Can Spec Publ Fish Aquat Sci:291.

[5] Raymond JA, Hassel A. Some characteristics of freezing avoidance in two osmerids, rainbow smelt and capelin. 2000. J Fish Biol;57(sa):1-7.

[6] Madigan DJ, Richardson AJ, Carlisle AB, Weber SB, Brown J, Hussey NE. Water column structure defines vertical habitat of twelve pelagic predators in the South Atlantic. 2021. ICES J Mar Sci;78(3):867-83.

[7] Sutton G, Pichegru L, Botha JA, Kouzani AZ, Adams S, Bost CA, et al. Multi-predator assemblages, dive type, bathymetry and sex influence foraging success and efficiency in African penguins. 2020. PeerJ;8:e9380.

[8] Brown A, Thatje S. The effects of changing climate on faunal depth distributions determine winners and losers. 2015. Glob Change Biol;21(1):173-80.

[9] Sallée JB, Pellichero V, Akhoudas C, Pauthenet E, Vignes L, Schmidtko S, et al. Summertime increases in upper-ocean stratification and mixed-layer depth. 2021. Nature;591(7851):592-8.

[10] Brill RW, Lutcavage ME. Understanding environmental influences on movements and depth distributions of tunas and billfishes can significantly improve population assessments. 2001. Am Fish Soc Symp;25:179-98.

[11] NMFS. Trawl rockfish conservation area (RCA) boundary modifications - final environmental assessment. 2014. NOAA Nat Mar Fish Serv Tech Rep:200.

[12] Bigelow KA, Maunder MN. Does habitat or depth influence catch rates of pelagic species? 2007. Can J Fish Aquat Sci;64(11):1581-94.

[13] Wright BM, Ford JKB, Ellis GM, Deecke VB, Shapiro AD, Battaile BC, et al. Fine-scale foraging movements by fish-eating killer whales (*Orcinus orca*) relate to the vertical distributions and escape responses of salmonid prey (*Oncorhynchus* spp.). 2017. Mov Ecol;5(1):3.

[14] Braun CD, Arostegui MC, Thorrold SR, Papastamatiou YP, Gaube P, Fontes J, et al. The functional and ecological significance of deep diving by large marine predators. 2022. Annu Rev Mar Sci;14:129-59.

[15] Maravelias CD. Habitat selection and clustering of a pelagic fish: Effects of topography and bathymetry on species dynamics. 1999. Can J Fish Aquat Sci;56:437-50.

[16] Genin A. Bio-physical coupling in the formation of zooplankton and fish aggregations over abrupt topographies. 2004. J Mach Learn;50:3-20.

[17] Borland HP, Gilby BL, Henderson CJ, Leon JX, Schlacher TA, Connolly RM, et al. The influence of seafloor terrain on fish and fisheries: A global synthesis. 2021. Fish Fish;22(4):707-34.

[18] Campana SE, Joyce W, Fowler M. Subtropical pupping ground for a cold-water shark. 2010. Can J Fish Aquat Sci;67(5):769-73.

[19] Logan RK, Vaudo JJ, Wetherbee BM, Shivji MS. Patrolling the border: Billfish exploit the hypoxic boundary created by the world's largest oxygen minimum zone. 2023. J Anim Ecol;92(8):1658-71.

[20] Hays GC. A review of the adaptive significance and ecosystem consequences of zooplankton diel vertical migrations. 2003. Hydrobiologia;503:163-70.

[21] Pothoven SA, Vanderploeg HA. Diet and prey selection of alewives in Lake Michigan: Seasonal, depth, and interannual patterns. 2004. Trans Am Fish Soc;133(5):1068-77.

[22] Kitagawa T, Kimura S, Nakata H, Yamada H. Diving behavior of immature, feeding Pacific bluefin tuna (*Thunnus thynnus orientalis*) in relation to season and area: The East China Sea and the Kuroshio-Oyashio transition region. 2004. Fish Oceanogr;13(3):161-80.

[23] Gallagher CP, Guzzo MM, Dick TA. Seasonal depth and temperature use, and diel movements of lake trout (*Salvelinus namaycush*) in a subarctic lake. 2019. Arctic Science;5(2):71-89.

[24] Afonso AS, Hazin FHV. Vertical movement patterns and ontogenetic niche expansion in the tiger shark, *Galeocerdo cuvier*. 2015. PLoS ONE;10(1):e0116720.

[25] Williams SM, Holmes BJ, Tracey SR, Pepperell JG, Domeier ML, Bennett MB. Environmental influences and ontogenetic differences in vertical habitat use of black marlin (*Istiompax indica*) in the southwestern Pacific. 2017. R Soc Open Sci;4(11):170694.

[26] Seitz AC, Norcross BL, Wilson D, Nielsen JL. Identifying spawning behavior in Pacific halibut, *Hippoglossus stenolepis*, using electronic tags. 2005. Environ Biol Fish;73(4):445-51.

[27] Teo SLH, Boustany A, Dewar H, Stokesbury MJW, Weng KC, Beemer S, et al. Annual migrations, diving behavior, and thermal biology of Atlantic bluefin tuna, *Thunnus thynnus< \i>, on their Gulf of Mexico breeding grounds.* 2007. Mar Biol;151(1):1-18.

[28] Werner EE, Gilliam JF, Hall DJ, Mittelbach GG. An experimental test of the effects of predation risk on habitat use in fish. 1983. Ecology;64(6):1540-8.

[29] Walters CJ, Juanes F. Recruitment limitation as a consequence of natural selection for use of restricted feeding habitats and predation risk taking by juvenile fish. 1993. Can J Fish Aquat Sci;50:2058-70.

[30] Braun CD, Gaube P, Sinclair-Taylor TH, Skomal GB, Thorrold SR. Mesoscale eddies release pelagic sharks from thermal constraints to foraging in the ocean twilight zone. 2019. Proc Natl Acad Sci USA;116(35):17187-92.

[31] Døving KB, Westerberg H, Johnsen PB. Role of olfaction in the behavioral and neuronal responses of Atlantic salmon, *Salmo salar* , to hydrographic stratification. 1985. Can J Fish Aquat Sci;42(10):1658-67.

[32] Dittman AH, Quinn TP. Homing in Pacific salmon: Mechanisms and ecological basis. 1996. J Exp Biol;199(1):83-91.

[33] Roscoe DW, Hinch SG, Cooke SJ, Patterson DA. Behaviour and thermal experience of adult sockeye salmon migrating through stratified lakes near spawning grounds: The roles of reproductive and energetic states. 2010. Ecol Freshw Fish;19(1):51-62.

[34] Crossin GT, Heupel MR, Holbrook CM, Hussey NE, Lowerre-Barbieri SK, Nguyen VM, et al. Acoustic telemetry and fisheries management. 2017. Ecol Appl;27(4):1031-49.

[35] Ford JK, Ellis GM, Barrett-Lennard LG, Morton AB, Palm RS, Balcomb III KC. Dietary specialization in two sympatric populations of killer whales (*Orcinus orca*) in coastal British Columbia and adjacent waters. 1998. Can J Fish Aquat Sci;76(8):1456-71.

[36] Shirk PL, Richerson K, Banks M, Tuttle V. Predicting bycatch of Chinook salmon in the Pacific hake fishery using spatiotemporal models. 2023. ICES J Mar Sci;80:133-44.

[37] Sabal MC, Richerson K, Moran P, Levi T, Tuttle VJ, Banks M. Warm oceans exacerbate Chinook salmon bycatch in the Pacific hake fishery driven by thermal and diel depth-use behaviours. 2023. Fish Fish.

[38] Orsi JA, Wertheimer AC. Marine vertical distribution of juvenile Chinook and coho salmon in southeastern Alaska. 1995. Trans Am Fish Soc;124(2):159-69.

[39] Hinke JT, Watters GM, Boehlert GW, Zedonis P. Ocean habitat use in autumn by Chinook salmon in coastal waters of Oregon and California. 2005. Mar Ecol Prog Ser;285:181-92.

[40] Walker RV, Myers KW. Behavior of Yukon River Chinook salmon in the Bering Sea as inferred from archival tag data. 2009. N Pac Anad Fish Com Bull;5:121-30.

[41] Smith JM, Fresh KL, Kagley AN, Quinn TP. Ultrasonic telemetry reveals seasonal variation in depth distribution and diel vertical migrations of sub-adult Chinook and coho salmon in Puget Sound. 2015. Mar Ecol Prog Ser;532:227-42.

[42] Arostegui MC, Essington TE, Quinn TP. Interpreting vertical movement behavior with holistic examination of depth distribution: A novel method reveals cryptic diel activity patterns of Chinook salmon in the Salish Sea. 2017. Anim Biotel;5(1):2.

[43] Courtney MB, Evans MD, Strøm JF, Rikardsen AH, Seitz AC. Behavior and thermal environment of Chinook salmon *Oncorhynchus tshawytscha* in the North Pacific Ocean, elucidated from pop-up satellite archival tags. 2019. Environ Biol Fish;102(8):1039-55.

[44] Healey MC. Life History of Chinook Salmon. In: Groot C, Margolis L, editors. Life Histories of Pacific Salmon. UBC Press; 1991. .

[45] Brodeur RD, Daly EA, Sturdevant MV, Miller TW, Moss JH, Thiess ME, et al. Regional comparisons of juvenile salmon feeding in coastal marine waters off the west coast of North America. 2007. Am Fish Soc Symp;57:183-203.

[46] Weitkamp LA. Marine distributions of Chinook salmon from the west coast of North America determined by coded wire tag recoveries. 2010. Trans Am Fish Soc;139(1):147-70.

[47] Freshwater C, Anderson SC, Beacham TD, Luedke W, Wor C, King J. An integrated model of seasonal changes in stock composition and abundance with an application to Chinook salmon. 2021. PeerJ;9:e11163.

[48] Crossin GT, Hinch SG. A nonlethal, rapid method for assessing the somatic energy content of migrating adult Pacific salmon. 2005. Trans Am Fish Soc;134:184-91.

[49] Beacham TD, Wallace C, MacConnachie C, Jonsen K, McIntosh B, Candy JR, et al. Population and individual identification of Chinook Salmon in British Columbia through parentage-based tagging and genetic stock identification with single nucleotide polymorphisms. 2018. Can J Fish Aquat Sci;75(7):1096-105.

[50] Center NNGD. British Columbia 3 arc-second Bathymetric Digital Elevation Model. NOAA National Centers for Environmental Information; 2007.

[51] Center NNGD. La Push, Washington 1/3 arc-second MHW Coastal Digital Elevation Model. NOAA National Centers for Environmental Information; 2007.

[52] Kelley D, Richards C. Oce: Analysis of oceanographic data; 2022.

[53] Thieurmel B, Elmarhraouii A. Suncalc: Compute sun position, sunlight phases, moon position and lunar phase; 2019.

[54] MacCready P, McCabe RM, Siedlecki SA, Lorenz M, Giddings SN, Bos J, et al. Estuarine circulation, mixing, and residence times in the Salish Sea. 2021. J Geophys Res Oceans;126(2):e2020JC016738.

[55] Haidvogel DB, Arango HG, Hedstrom K, Beckmann A, Malanotte-Rizzoli P, Shchepetkin AF. Model evaluation experiments in the North Atlantic Basin: Simulations in nonlinear terrain-following coordinates. 2000. Dyn Atmo Ocean;32(3-4):239-81.

[56] Shchepetkin AF, McWilliams JC. The regional oceanic modeling system (ROMS): A split-explicit, free-surface, topography-following-coordinate oceanic model. 2005. Ocean Mod;9(4):347-404.

[57] Winslow L, Read J, Woolway R, Brentrup J, Leach T, Zwart J, et al.. rLakeAnalyzer: Lake physics tools; 2019.

[58] Lerner JE, Hunt BPV. Seasonal variation in the lipid content of Fraser River Chinook Salmon (*Oncorhynchus tshawytscha*) and its implications for Southern Resident Killer Whale (*Orcinus orca*) prey quality. 2023. Sci Rep;13:2675.

[59] Breiman L. Random forests. 2001. Mach Learn;45(1):5-32.

[60] Friedman JH. Greedy function approximation: A gradient boosting machine. 2001. Ann Statist;29(5).

[61] Kuhn M. Caret: Classification and regression training; 2022.

[62] Wright MN, Ziegler A. Ranger: A fast implementation of random forests for high dimensional data in C++ and R. 2017. J Stat Soft;77(1):1-17.

[63] Wager S, Hastie T, Efron B. Confidence intervals for random forests: The jackknife and the infinitesimal jackknife. 2015. Journal of Machine Learning Research;15(1):1625-51.

[64] R Core Team. R: A Language and Environment for Statistical Computing. Vienna, Austria: R Foundation for Statistical Computing; 2021.

[65] Geurts P, Ernst D, Wehenkel L. Extremely randomized trees. 2006. Mach Learn;63(1):3-42.

[66] Bakun A. Vertical ambush corridors: Intriguing multi-mechanism ecological structures embedded in the kinetic fluid architectures of ocean living resource production systems. 2023. Fish Fish;24(1):3-20.

[67] Bakun A. Patterns in the Ocean: Processes and Marine Population Dynamics. La Jolla, California: California Sea Grant; 1996.

[68] Pike GC. Age, Growth and Maturity Studies on the Pacific Anchovy (*Engraulis Mordax*) from the Coast of British Columbia [Master of Arts]. Univeristy of British Columbia; 1951.

[69] Carlson HR. Seasonal distribution and environment of Pacific herring near Auke Bay, Lynn Canal, Southeastern Alaska. 1980. Trans Am Fish Soc;109:71-8.

[70] Brown ED, Seitz J, Norcross BL, Huntington HP. Ecology of herring and other forage fish as recorded by resource users of Prince William Sound and the outer Kenai Peninsula, Alaska. 2002. Ala Fish Res Bull;9(2):75-101.

[71] Hanson MB, Baird RW, Ford JKB, Hempelmann-halos J, Doornik DMV, Candy JR, et al. Species and stock identification of prey consumed by endangered southern resident killer whales in their summer range. 2010. Endanger Species Res;11:69-82.

[72] Hanson MB, Emmons CK, Ford MJ, Everett M, Parsons K, Park LK, et al. Endangered predators and endangered prey: Seasonal diet of southern resident killer whales. 2021. PLoS ONE;16(3):e0247031.

[73] Thornton SJ, Toews S, Stredulinsky EH, Gavrilchuk K, Burnham R, Noren DP, et al. Southern resident killer whale (*Orcinus orca*) summer distribution and habitat use in the southern Salish Sea and the Swiftsure Bank area (2009 to 2020). 2022. CSAS Res Doc;2022/037:56 p.

[74] Gregr EJ, Trites AW. Predictions of critical habitat for five whale species in the waters of coastal British Columbia. 2001. Can J Fish Aquat Sci;58(7):1265-85.

[75] Gregr E, Trites A. A novel presence-only validation technique for improved Steller sea lion Eumetopias jubatus critical habitat descriptions. 2008. Mar Ecol Prog Ser;365:247-61.

[76] Carlisle AB, Perle CR, Goldman KJ, Block BA. Seasonal changes in depth distribution of salmon sharks (*Lamna ditropis*) in Alaskan waters: Implications for foraging ecology. 2011. Can J Fish Aquat Sci;68(11):1905-21.

[77] Berman CH, Quinn TP. Behavioural thermoregulation and homing by spring Chinook salmon, *Oncorhynchus tshawytscha* (Walbaum), in the Yakima River. 1991. J Fish Biol;39(3):301-12.

[78] Minke-Martin V, Hinch SG, Braun DC, Burnett NJ, Casselman MT, Eliason EJ, et al. Physiological condition and migratory experience affect fitness-related outcomes in adult female sockeye salmon. 2018. Ecol Freshw Fish;27(1):296-309.

[79] Hinke JT, Foley DG, Wilson C, Watters GM. Persistent habitat use by Chinook salmon *Oncorhynchus tshawytscha* in the coastal ocean. 2005. Mar Ecol Prog Ser;304:207-20.

[80] Winant CD, Bratkovich AW. Temperature and currents on the southern California shelf: A description of the variability. 1981. J Phys Oceanogr;11(1):71-86.

[81] Hay DE, McCarter PB. Continental shelf area and distribution, abundance, and habitat of herring in the North Pacific. In: Forage Fishes in Marine Ecosystems: Proceedings of the International Symposium on the Role of Forage Fishes in Marine Ecosystems. vol. 97–01. Anchorage, Alaska, USA: University of Alaska Sea Grant College Program, Fairbanks, Alaska; 1996. p. 559-72.

[82] Ahrens RNM, Walters CJ, Christensen V. Foraging arena theory. 2012. Fish Fish;13(1):41-59.

[83] Loher T, Webster RA, Carlile D. A test of the detection range of acoustic transmitters and receivers deployed in deep waters of Southeast Alaska, USA. 2017. Anim Biotel;5(1):27.

[84] Holbrook C, Hayden T, Binder T, Pye J. Glatos: A package for the Great Lakes acoustic telemetry observation system; 2021.

[85] Metzger EJ, Smedstad OM, Thoppil P, Hurlburt H, Cummings J, Walcraft A, et al. US Navy operational global ocean and Arctic ice prediction systems. 2014. Oceanog;27(3):32-43.

[86] Mass CF, Albright M, Ovens D, Steed R, Maciver M, Grimit E, et al. Regional environmental prediction over the Pacific northwest. 2003. Bull Amer Meteor Soc;84(10):1353-66.

[87] Davis KA, Banas NS, Giddings SN, Siedlecki SA, MacCready P, Lessard EJ, et al. Estuary-enhanced upwelling of marine nutrients fuels coastal productivity in the U.S. Pacific Northwest. 2014. J Geophys Res Oceans;119(12):8778-99.

[88] Siedlecki SA, Banas NS, Davis KA, Giddings S, Hickey BM, MacCready P, et al. Seasonal and interannual oxygen variability on the Washington and Oregon continental shelves. 2015. J Geophys Res Oceans;120(2):608-33.

[89] Hoyer S, Hamman J. Xarray: N-d labeled arrays and datasets in Python. 2017. J Open Res Softw;5(1):10.

[90] Kowarik A, Templ M. Imputation with the R Package VIM. 2016. J Stat Soft;74(7):1-16.

[91] Bürkner PC. Brms: An R package for Bayesian multilevel models using Stan. 2017. J Stat Soft;80(1):1-28.

Supplementary Figures

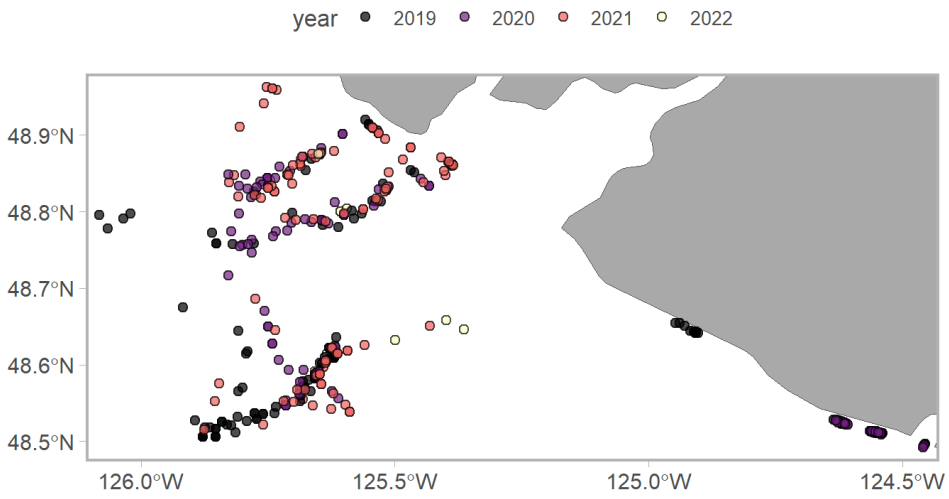


Figure S1: Locations of tag releases. See Figure 1 for additional information.

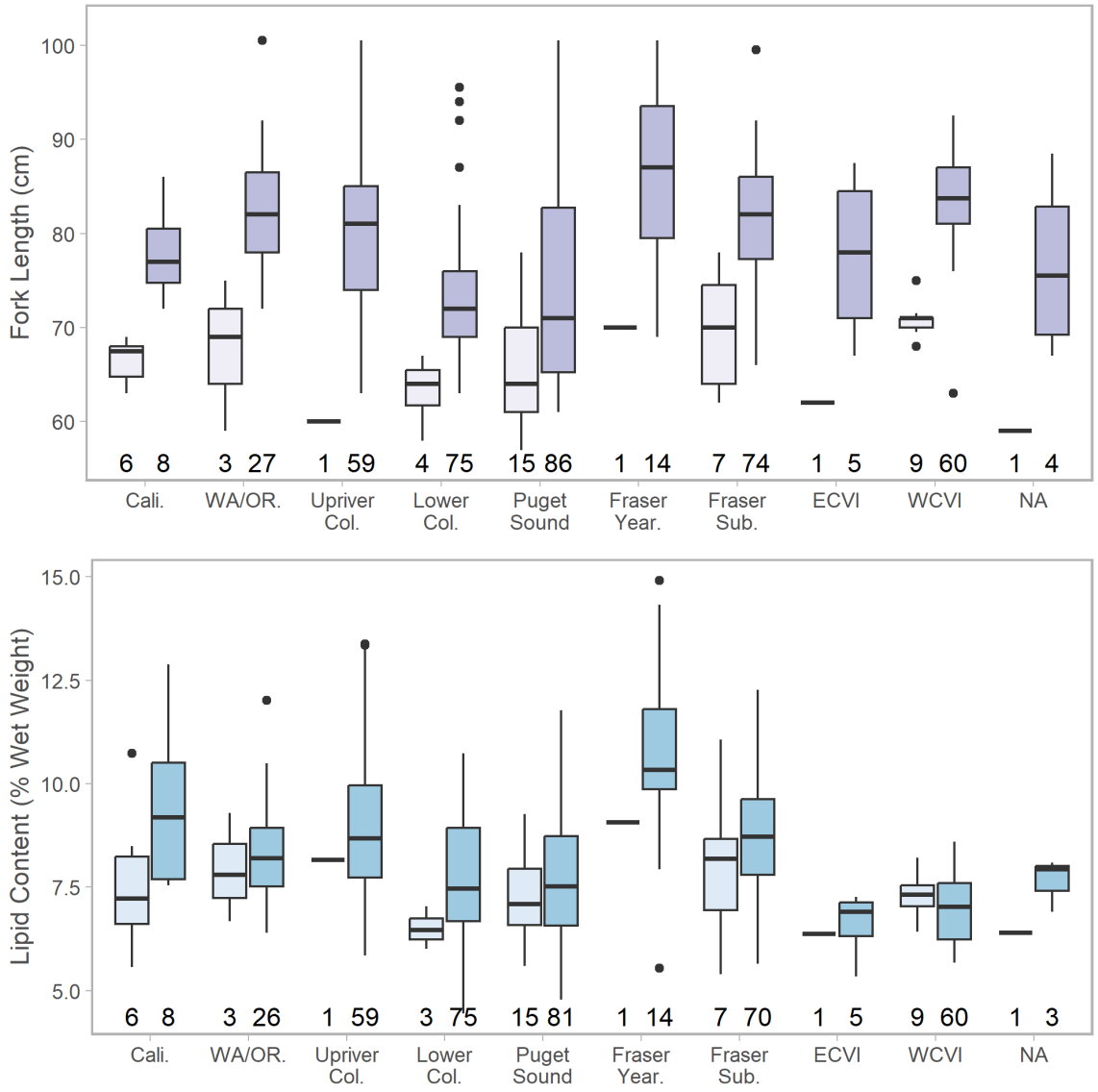


Figure S2: Chinook salmon fork length (top) and estimated whole body lipid content (bottom). Immature (mature) individuals are represented by lighter (darker) colours. Stock aggregate- and stage-specific sample sizes are shown as numerics under each boxplot. Stock aggregate abbreviations and contributing stocks are defined in Table S1.

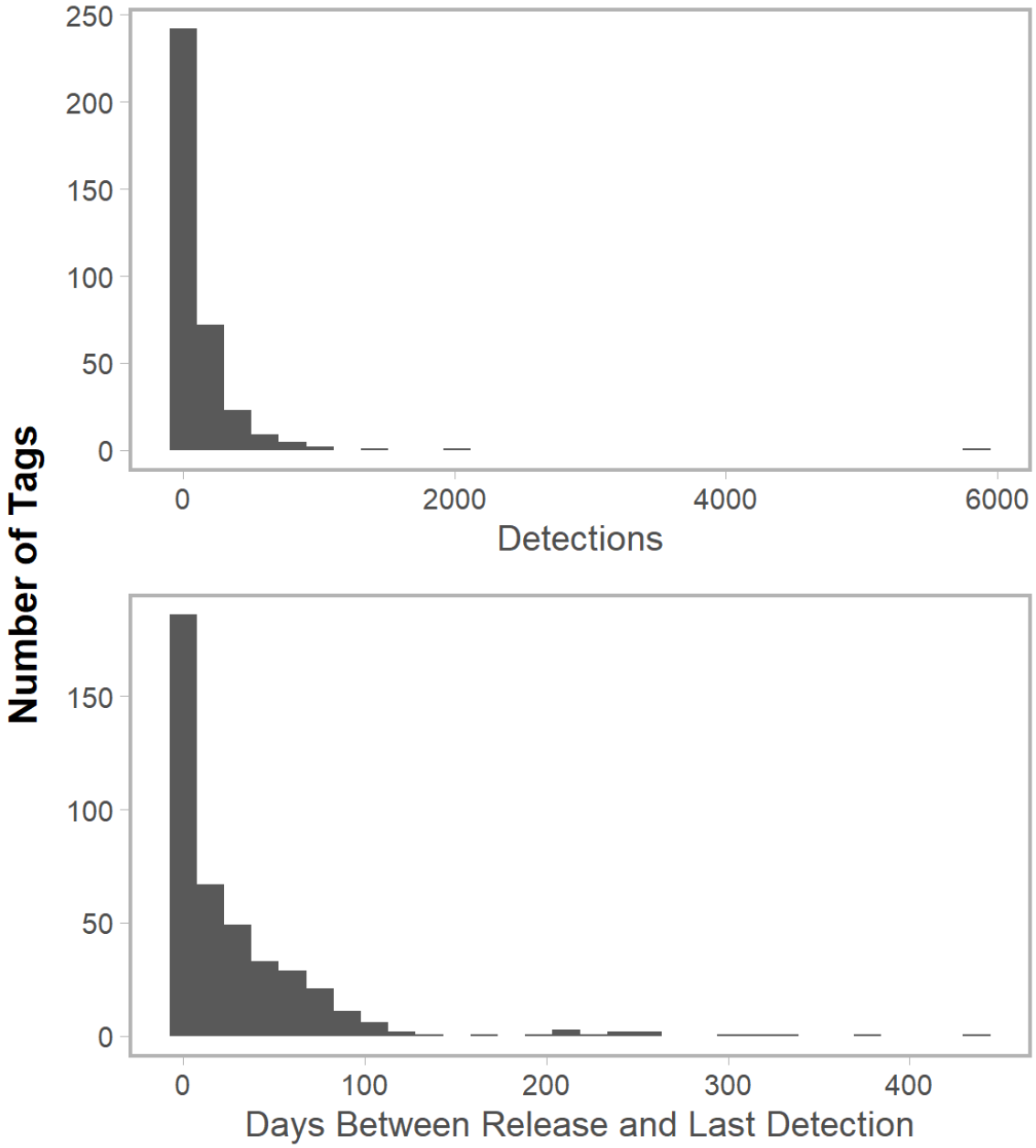


Figure S3: Number of detections (top) and number of days between release and last detection (bottom) per tag.

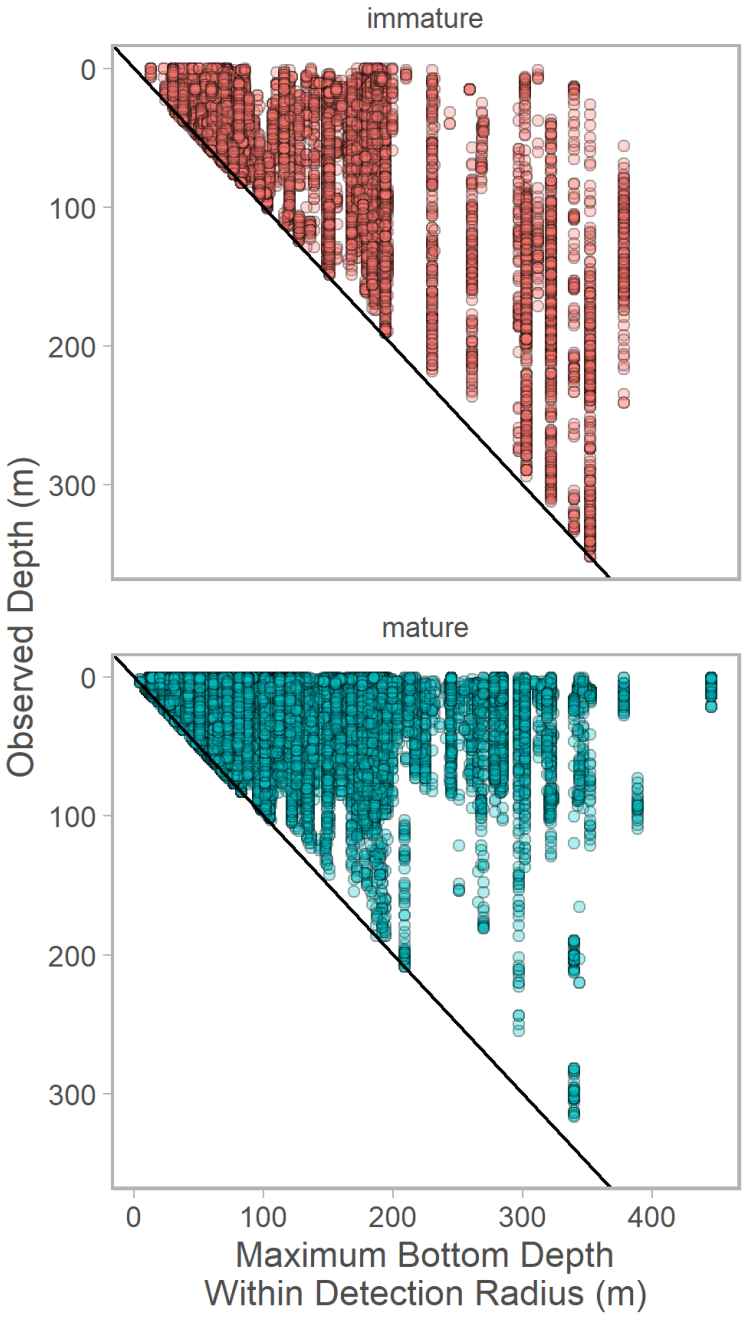


Figure S4: Observed depth relative to bottom depth (maximum within a receiver's detection radius). Black line represents the ocean bottom.

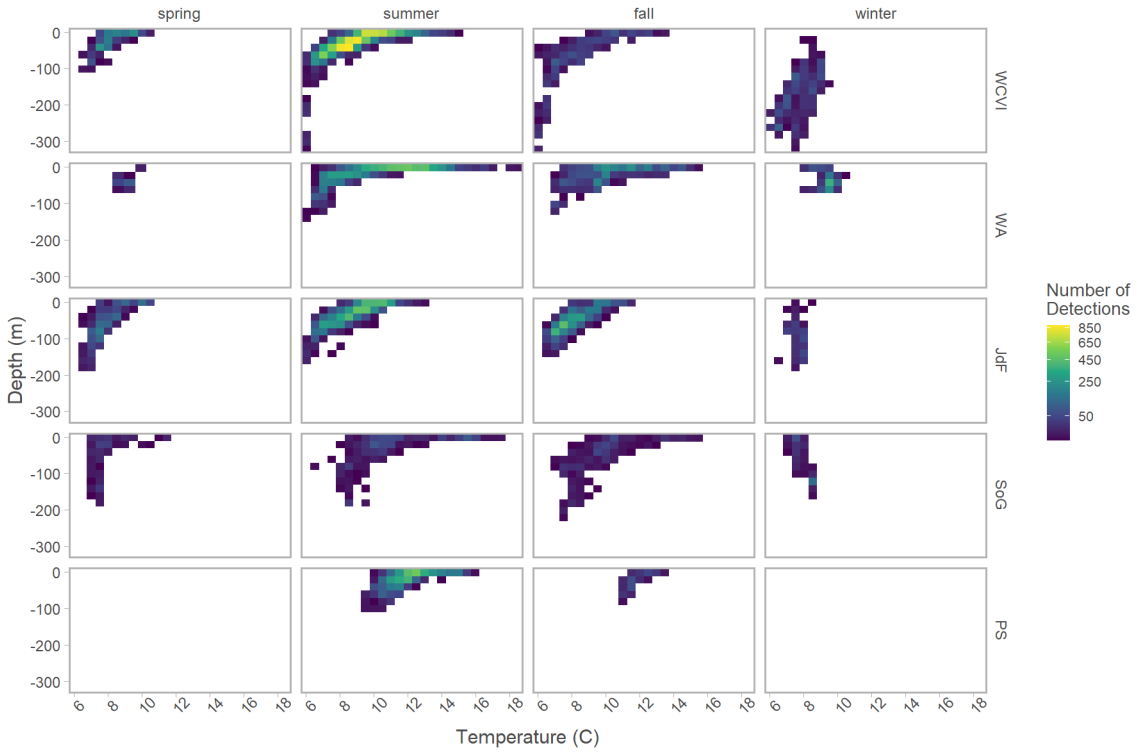


Figure S5: Seasonal and spatial variation in temperature-at-depth coincident with Chinook salmon detections. Seasons defined as: winter (December-February), spring (March-May), summer (June-August), and fall (September-November).

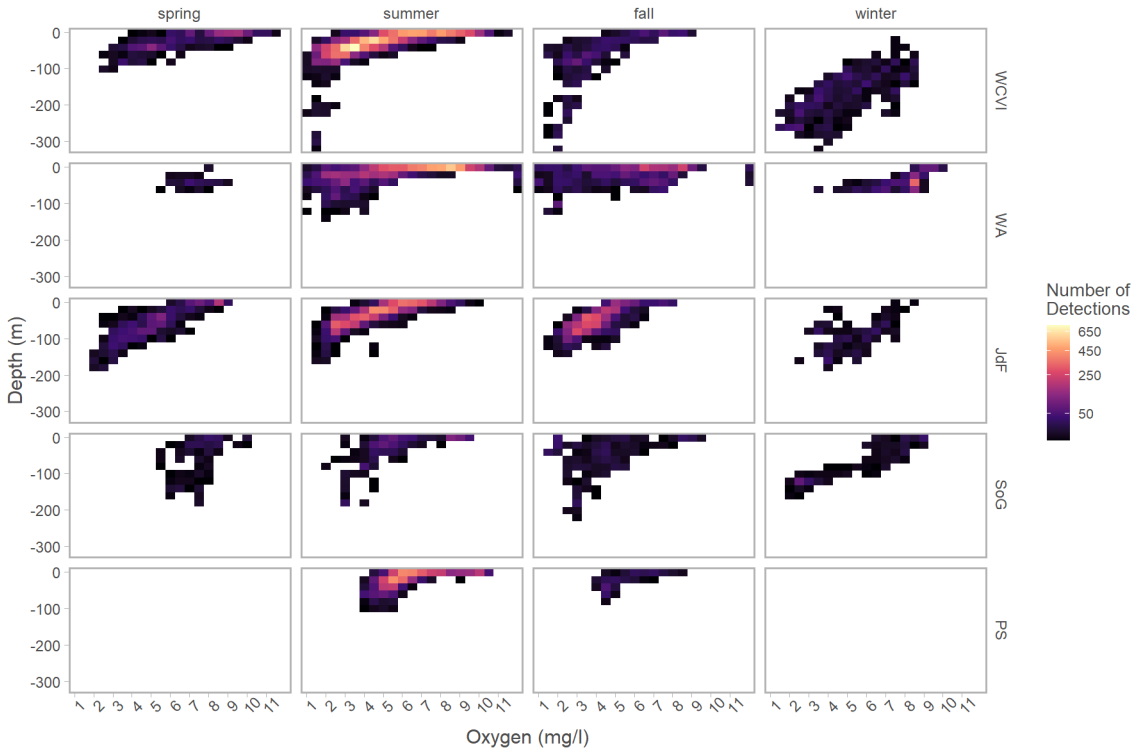


Figure S6: Seasonal and spatial variation in dissolved oxygen concentration (mg/l) coincident with Chinook salmon detections. Seasons defined as: winter (December-February), spring (March-May), summer (June-August), and fall (September-November).

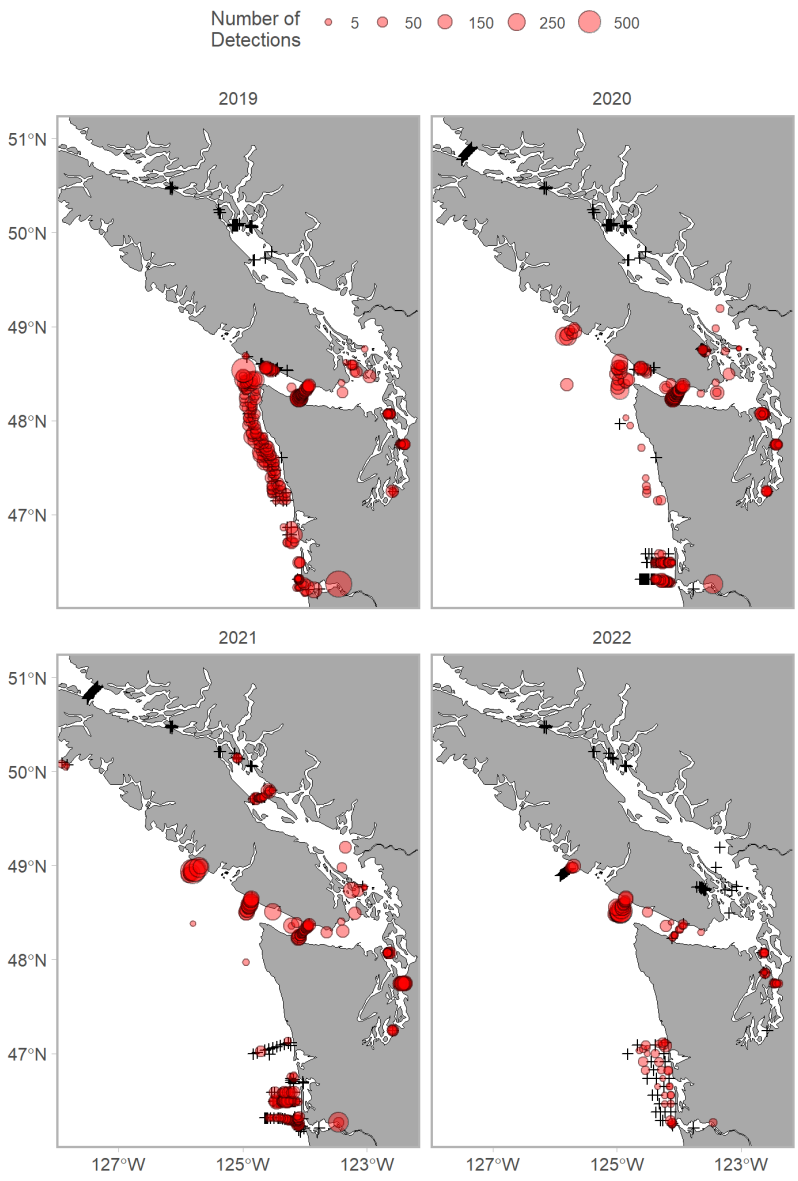


Figure S7: Number of individual detections at each acoustic receiver. Locations that were not used in model fitting (no ROMS data available or no detections) are shown as crosses.

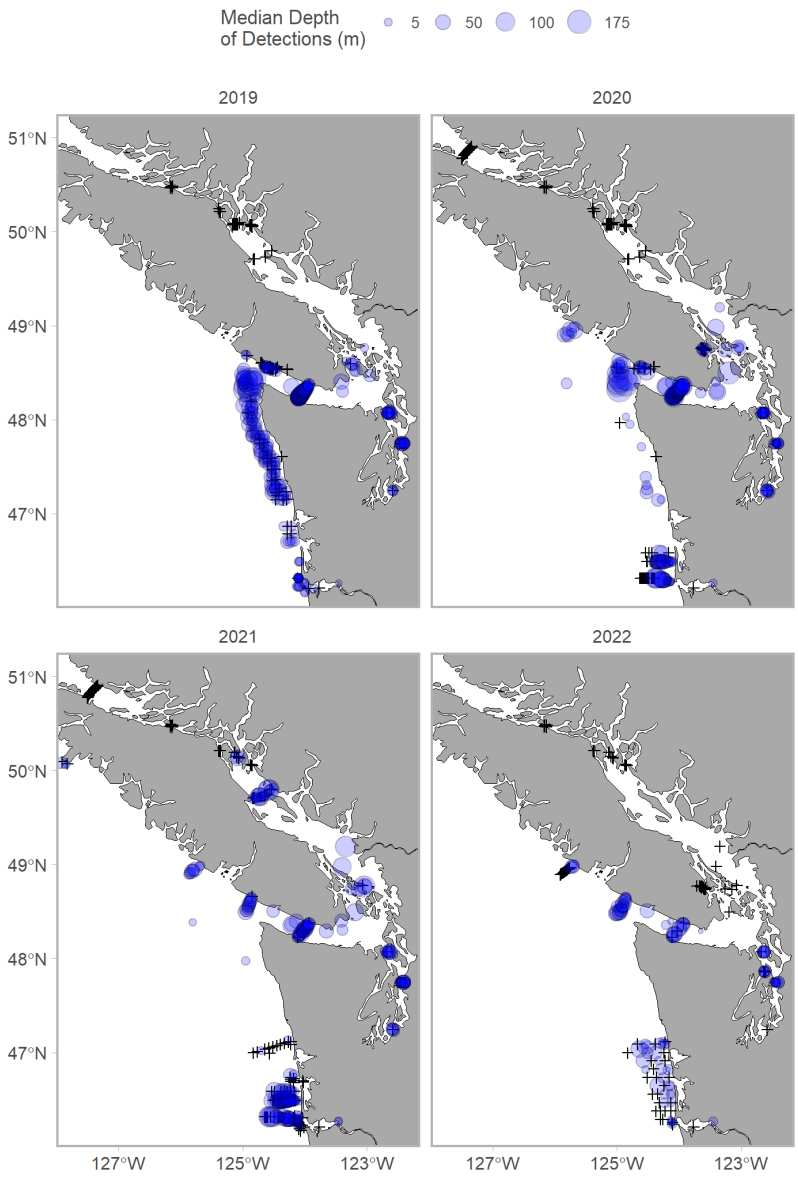


Figure S8: Mean depth of detections at each acoustic receiver. Locations that were not used in model fitting (no ROMS data available or no detections) are shown as crosses.

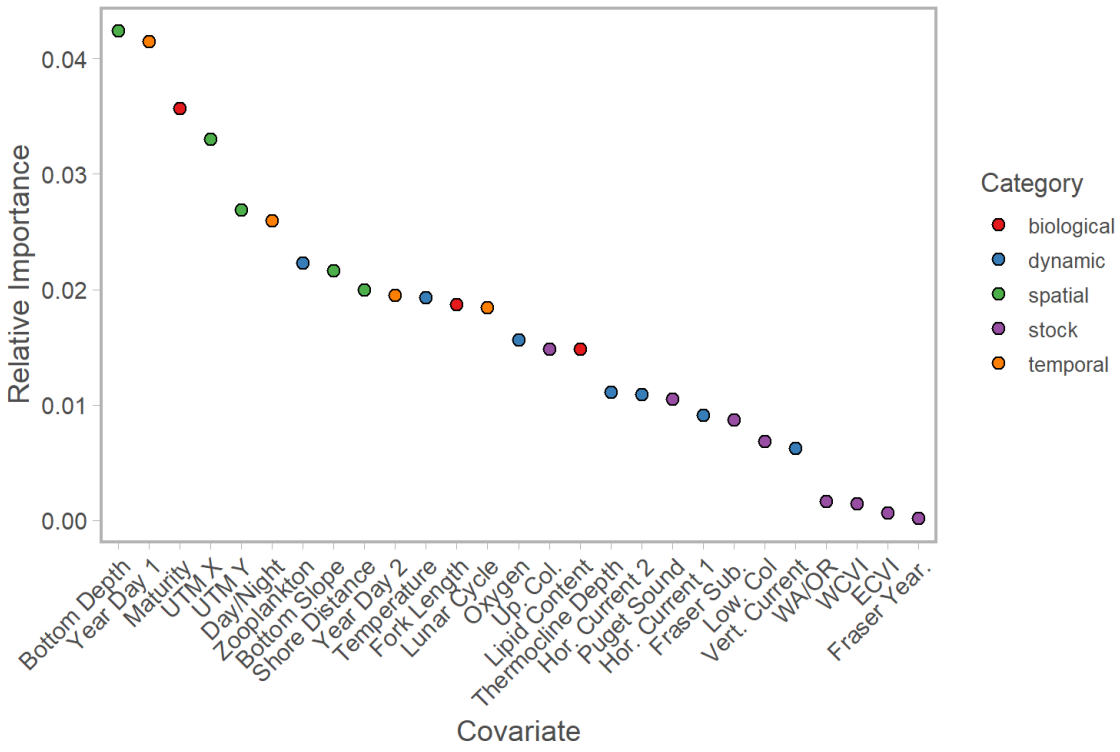


Figure S9: Relative importance of explanatory variables in random forest model including stock aggregate identity (equivalent to Figure 3). Importance is quantified as the difference (mean among trees) in root mean square error when a given variable is excluded during out-of-bag prediction. An uncorrelated random variable will have a difference in root mean square error of zero. Colors represent categories of explanatory variables (note the purple 'stock' categories at the right).

Supplementary Methods

Detections Data and Preprocessing

It was not feasible to conduct a comprehensive range test given the spatial scope and seasonal variation in environmental conditions; however, 800 m is a conservative estimate of the detection radius of tags used here [83]. Receivers record the unique tag code, a detection timestamp, and an estimate of depth (precision +/- 10 m). Detections and receiver metadata were either provided directly by the scientists managing the arrays or by the Ocean Tracking Network (https://oceantrackingnetwork.org/). Prior to statistical analyses, we removed false detections using the glatos R package (< 1%) of detections assuming a short interval criteria of 3600 seconds [84] or more than 2.5 m below the maximum bottom depth within a receiver's detection radius. We fixed depth estimates less than 2.5 m below the maximum bottom depth (< 1.0% of detections) at 0.5 m above the maximum bottom depth value and estimates less than 0.1 m below the surface (i.e., at surface or slightly above surface) at 0.1 m below the surface to allow for logit transformations.

Seasonal Effects

We converted calendar day to two composite variables that cumulatively represent a cyclical process,

$$d_1 = \cos\left(2\pi \frac{d}{365}\right), \tag{1a}$$

$$d_2 = \sin\left(2\pi \frac{d}{365}\right), \tag{1b}$$

where d is calendar day. We note that the use of composite variables involves trade-offs. When a single attribute (i.e., calendar day) is split into two covariates, its weight relative to other covariates is inflated. Additionally the split rules associated with machine learning algorithms do not process linked covariates simultaneously, resulting

in a subset of regression trees containing d_1, but not d_2, or vice versa. Nevertheless, we decided that an imperfect cyclical representation was better than alternative methods of incorporating seasonal variability that would assume independent or linear relationships through time.

LiveOcean ROMS Model

The LiveOcean implementation applies ROMS to a model domain that includes all of Oregon, Washington, the Vancouver Island coastal waters, and the Salish Sea. In the horizontal plane, the model grid follows lines of constant latitude and longitude, with higher resolution (500 m) in most of the Salish Sea and Washington coastal estuaries and lower resolution (up to 3 km) in the open ocean. The grid is terrain-following in the vertical direction, with 30 levels following the bathymetry and free surface, and with higher resolution near the sea surface and bed.

Model forcing includes open ocean boundary conditions from a global ocean model [85]; atmospheric forcing from a regional weather model [86]; tidal forcing; and daily averaged flow from 45 rivers. In addition, LiveOcean includes a nitrogen, phytoplankton, microzooplankton, detritus, and oxygen (NPZDO) biogeochemical model [87, 88].

LiveOcean is run in a data assimilative mode to generate daily forecasts, from which we extracted data at specified spatiotemporal coordinates (northing, easting, depth, date, and hour of tag detections) using linear interpolation routines in the xarray Python module [89].

Data Imputation

Before fitting models, we imputed missing values for stations where some, but not all, ROMS variables were missing due to boundary conditions (≈ 1-10% of observations) and several fish that were missing lipid content estimates (due to equipment

malfunction; 2% of observations). Stations with no ROMS data were excluded. We used k-Nearest Neighbor Imputation, using the R package VIM [90], to impute all variables. ROMS variables were imputed based on spatial location, date, distance to shore, mean bottom depth, and mean slope. Full body lipid content was imputed based on fork length, stock identity (as inferred from genetic stock identification via single nucleotide polymorphisms [49]), and maturity stage.

Life History Stage Model

The maturity stage at the time of tagging could not be inferred for a subset of individuals because they were not detected in terminal areas due to mortality or because detection efficiency in their natal system was very low. To predict the maturity stage of "unknown" individuals, we used data from 291 individuals where maturation stage at the time of tagging could be identified post-hoc based on detection histories.

Specifically we fit a hierarchical generalized linear model that modelled the probability of maturity using a Bernoulli distribution and a logit link:

$$y_i \sim \text{Bernoulli}(\mu_i), \tag{1}$$

$$\text{logit}(\mu_i) = \beta_d d_i + \beta_l l_i + \alpha_s, \tag{2}$$

$$\alpha_s \sim \text{Normal}\left(\mu_{\alpha_s}, \sigma^2_{\alpha_s}\right), \tag{3}$$

where y_i represents the binomial maturation stage at the time of tagging (0 immature, 1 mature) for individual i and μ is the mean probability of maturity. The β parameters represent slope estimates for tagging date (year day) d and fork length l, while α_s represents a random intercept for stock identity.

We fit the model in a Bayesian framework with weakly informative priors (μ_{α_s} = Normal $(0, 10)$, β = Normal $(0, 2)$ and σ_{α_s} = Cauchy $(0, 1)$) in the R package brms [91]. The model was fit with four chains each with 1500 iterations, 750 of which were dis-

carded as warmup. Posterior checks indicated the chains mixed well and all model parameters had an effective sample size greater than 900 and an $\hat{R}$ less than 1.01, suggesting convergence.

The probability of maturity at the time of tagging was correlated with tagging date and fork length ($\beta_d = -0.08$ and $\beta_l = 0.32$) with significant variability among populations ($\sigma_s = 1.63$). As a result, larger fish and those tagged early in the year were the most likely to mature in the year they were tagged Figure S10. The fitted model had an 87% correct classification accuracy, with a classication threshold of 50%, within the training dataset.

We then used the estimated model to predict maturation probability for individuals with an "unknown" maturation stage based on detections data. If the 90th percentile interval of the posterior exceeded 50% for a given individual then they were assigned a mature classification for remaining analyses.

Machine Learning Model Tuning and Selection

To identify the framework with the best out-of-sample predictive performance we fit six alternative models within two machine learning frameworks (gradient boosting machines (GBM) and random forests (RF)) and three different response variable distributions (untransformed depth, depth scaled by maximum bottom bathymetry (bathymetric depth ratio), and logit transformed bathymetric depth ratio). For each of the six models we evaluated all combinations of hyperparameters via a grid search tuning procedure (Table S1) and selected the optimal suite of hyperparameters based on root mean square error (RMSE). We used two different random forest split rules to evaluate both traditional random forest and extremely randomized trees, which differ by being fit to the entire dataset (not a bootstrap sample) and where split points within a covariate are identified randomly following a uniform distribution [65].

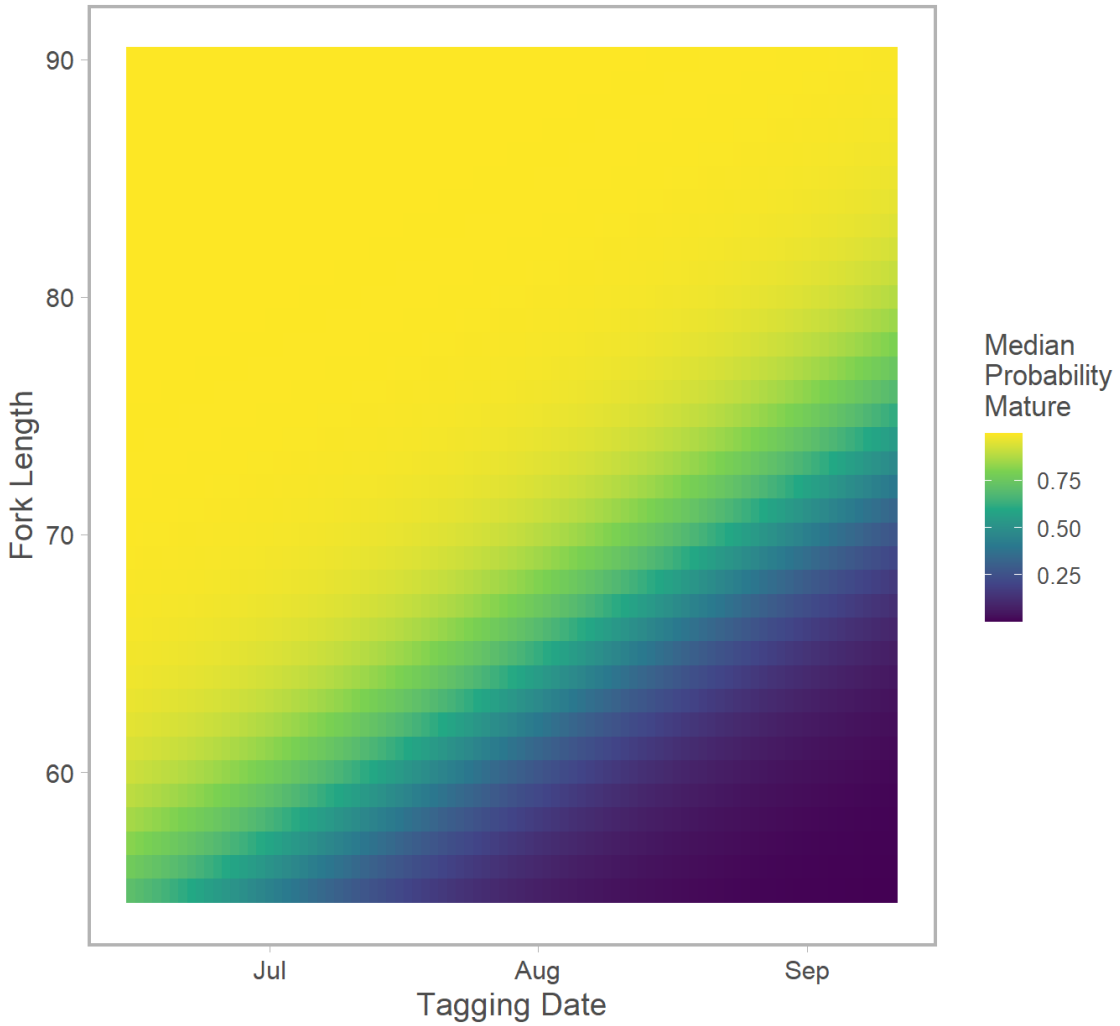


Figure S10: Median posterior predicted probability of maturity at time of tagging as a function of fork length and tagging date.

We next fit each model, fixing hyperparameters at their respective optimal values (lowest RMSE; Figures S11, S12), to both the training and out-of-sample testing datasets (described in methods above), then calculated the RMSE of predictions. In the case of transformed response variables (bathymetric depth ratio and logit-scaled bathymetric depth ratio), we backtransformed predictions and calculated RMSE relative to untransformed observations.

Random forest models tended to outperform gradient boosting machines, though these differences were relatively modest when predictions were evaluated using held-

Table S1: Model hyperparameters.

Model	Hyperparameter	Values
Gradient Boosting Machine	Interaction Depth	2, 5, 10
	Number of Trees	10, 20, 50, 100, 150, 200, 250, 300
	Shrinkage	0.01, 0.1
	Minimum Number of Observations in a Node	5, 10, 20
Random Forest	Number of Parameters at Split	2, 3, 5, 7, 9, 11, 13, 15, 17, 19
	Split Rule	variance, extratrees (extremely random trees)
	Number of Trees	500, 1000, 1500, 2000, 2500

out testing data. While the depth random forest model had the best performance with training data, the relative depth random forest model had the best performance with held-out testing data (Figure S13). Therefore we selected a random forest model fit to bathymetric depth ratio, with 1500 trees, and that evaluated 13 variables at each split for our primary analysis (Figure S13).

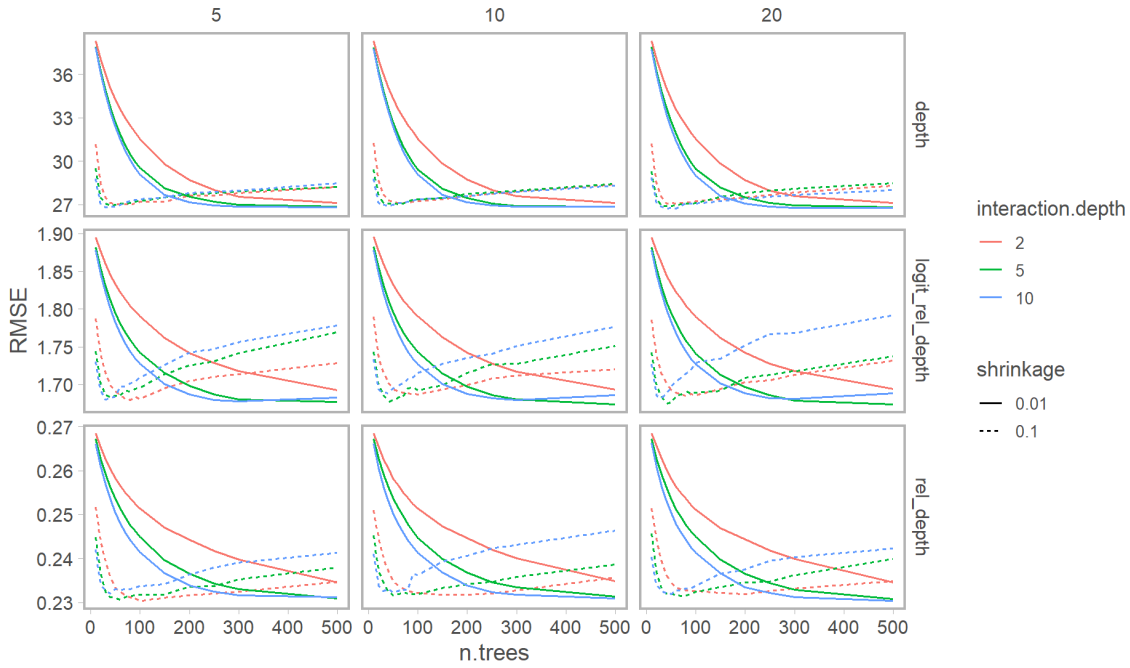


Figure S11: Performance of alternative gradient boosting machine model hyperparamters fit to training data. The columns represent different minimum observations per node. Root mean square error was calculated in transformed space (i.e., bathymetric depth ratio and logit-scaled bathymetric depth ratio were not back-transformed), so difference in scale among response variable types differs (figure rows).

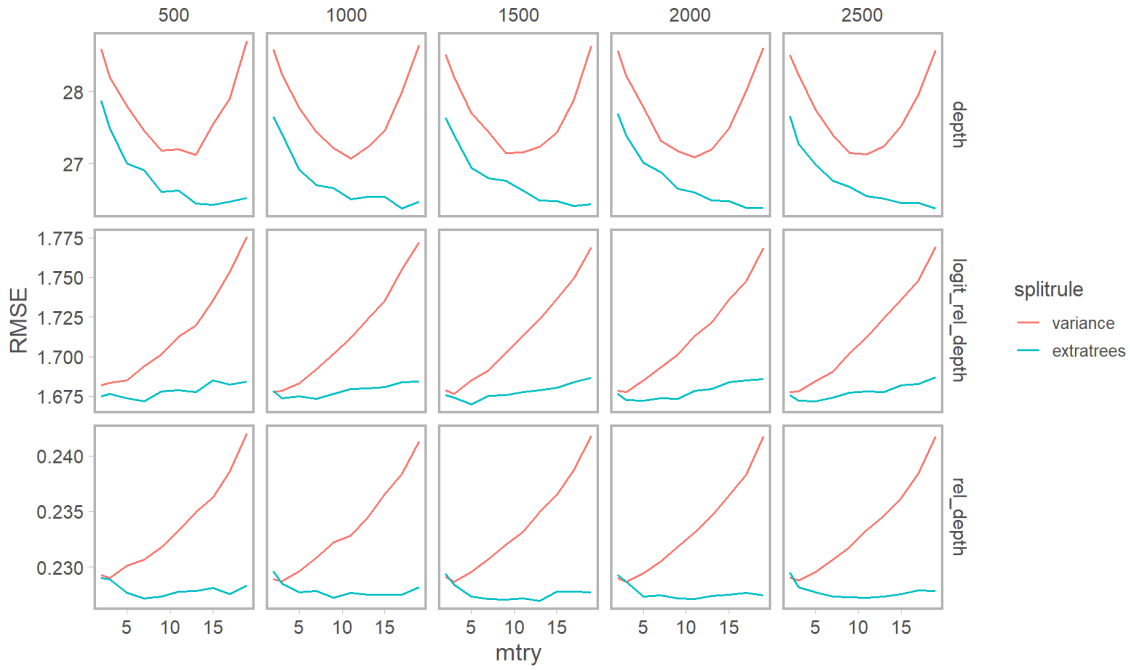


Figure S12: Performance of alternative random forest model hyperparamters fit to training data. Root mean square error was calculated in transformed space (i.e., bathymetric depth ratio and logit-scaled bathymetric depth ratio were not back-transformed), so difference in scale among response variable types differs (figure rows).

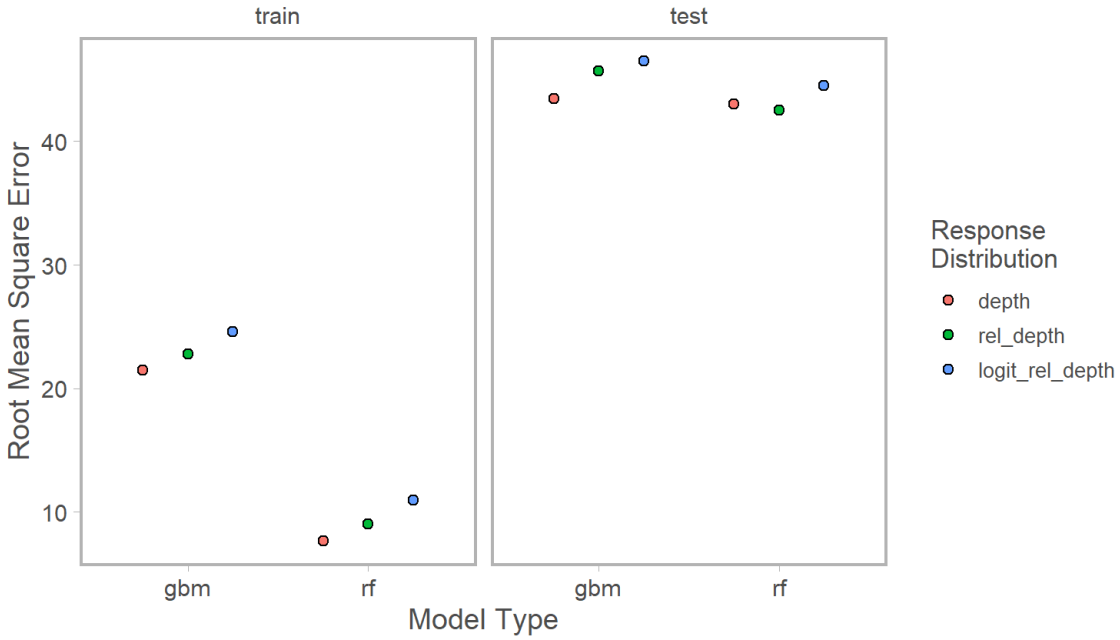


Figure S13: Predictive performance of six candidate machine learning models relative to testing (left) and out-of-sample training data (right). Root mean square error was calculated in real space (i.e., bathymetric depth ratio and logit-scaled bathymetric depth ratio were back-transformed).